

B3: Positivity Certificates for Counting Sequences

Three complete proofs, structural partial results, and exact reproducible diagnostics

Research prepared for the attached B3 program

6 October 2026

Contents

Findings and scope	2
Question register	2
How to read the certificates	4
Contents of the accompanying package	4
A sharp imaginary-part bound for Stanley-nonnegative polynomials	5
Result and exact scope	5
A centered interval lemma	5
Proof of the theorem	5
Stronger result: the complete nonreal root locus	7
Low-degree controls	8
Literature verification and novelty qualification	8
Q165: an all-length coefficient certificate and a Hurwitz-stability proof	9
Statement and the key positive polynomial	9
An exact rational generating function	9
Change variables to expose a positive recurrence	10
The four initial series and positivity	11
From the coefficient certificate to Hurwitz stability	12
Resolution of Q165	13
Exact verification and sources	13
Q166: A four-functions proof of log-concavity	14
General interval-constraint theorem	14
Application to forbidden runs of one-parts	15
Application to Q164: ordinary log-concavity for path powers	16
Sources and verification	16
Higher-order Stirling positivity: structural results and exact certificates	18
Results and scope	18
Definitions	19
Every higher-order Stirling triangle is TN of order two	19
Extreme coefficients of every cycle Hankel minor are positive	21
An all-parameter subset Hankel countercertificate	23
Why the direct production-matrix route fails	24
Exact computations and reusable certificates	25
Next proof targets	26
Interpolated totally positive sequences and multiplier tests	27
A compact exact parameter space for the interpolation questions	27

Complete degree-two analysis	27
A proof that $R_3=3$	28
Infinite families giving stronger lower bounds for r_n	30
Exact real-root gap witnesses	30
Q1249: a rigorous obstruction to a tempting weakening	32
Reproduction and limitations	32
Q167: Complete classification in degrees at most four	34
Classification and explicit witnesses	34
Preliminary inequalities	34
Exhaustiveness for $n=1,2,3$	35
The degree-four obstruction	35
General obstruction available in every degree	35
Sources	35
Eulerian recognition, atoms, pattern intervals, and random graphs	37
Q1262: a finite recognition theorem for every possible length	37
Q1263: exact symmetric decompositions	39
Q2212: exhaustive finite coverage of every numerical monoid	39
Q1366: exact classical-pattern interval controls	40
Q2400: a rigorous growing prefix with high probability	40
Q183 and Q1454: structural calibrations and remaining gaps	41
Sources	42
Exact positivity diagnostic and verification	43
Supported inputs	43
Boundary-safe exact methods	43
Independent verification	44
Most useful remaining proof targets	44

Findings and scope

Three of the twenty exact questions have complete proofs: Q165, Q166 and Q325. The proofs include stronger statements: strict left-half-plane location of all nonzero path-domination roots; weighted log-concavity for binary words under arbitrary interval-sum constraints; and equality cases plus the full attainable root locus for Stanley-nonnegative polynomials.

The remaining work gives several all-parameter partial theorems, complete small-degree classifications, exact recognition procedures, and finite certificates. The main additional results are total nonnegativity of order two for both higher-order Stirling triangles for every order parameter; positivity of both extreme coefficients of every cycle Hankel minor; exact interpolation extrema in degrees two and three; an infinite Pell family of s -Eulerian representations; and a growing log-concave prefix for sparse random-graph independence counts with an explicit probability bound.

These are research derivations from the user's 6 October 2026 handoff. The central proofs were independently checked within this research run. Source statements and relevant prior results were checked against primary sources. No assertion of publication priority or exhaustive literature coverage is made. Complete proofs, rather than finite computational agreement, support the three completed statements.

Question register

Question	Strongest result established	Full target
Q164	Weighted ordinary log-concavity for every path power	ULC and full cycle case remain
Q165	Both parity polynomials real-rooted; stronger strict Hurwitz theorem after removing zero	Proved
Q166	Weighted interval-constraint theorem proves every forbidden-run composition case	Proved
Q167	Complete integer-factor coefficient orders through degree four	General degree remains
Q183	Real-rootedness for boards with at most two rows or columns	General avoidance class remains
Q325	Sharp imaginary bound, equality, and attainable root locus	Proved
Q1158	All minors of order at most two, every $r \geq 1$	Orders at least three remain
Q1159	All minors of order at most two, every $r \geq 1$	Orders at least three remain
Q1160	Highest and lowest nonzero coefficient of every Hankel minor positive	Interior coefficients remain
Q1161	Exact strict Hurwitz certificates after removing zero, $1 \leq n \leq 80$	All lengths remain
Q1249	Exact bounded tests and a precise obstruction to weakening the input class	Preserver characterization remains
Q1262	All-length recognition, exact exclusions, infinite Pell family	General structural classification remains
Q1263	Exact decomposition and degree-two gamma formulas	General interpretation remains
Q1366	Every interval with top length at most seven passes	All intervals remain
Q1441	$r_2 = (1 + \sqrt{2})/2$; lower bounds $5/4$ and $51/40$ in infinite degree families	Uniform upper bound remains
Q1442	$R_2 = 2$, $R_3 = 3$; full degree-two root locus	Degrees at least four remain
Q1443	Exact example forces every universal gap bound to be at least $\sqrt{13/5}$	Bound below two remains
Q1454	Explicit measures and exact checks for known calibration cases	General pattern class remains
Q2212	Exhaustive over every numerical monoid for all cutoffs through 22	Larger cutoffs remain

Question	Strongest result established	Full target
Q2400	Simultaneous prefix of order $n^{1/3}$ with exponentially high probability	Complete row remains

The machine-readable ledger preserves all twenty original statement IDs and records each scope and limitation separately. None of the twenty exact questions was refuted. A uniform negative coefficient theorem for the related **subset** Hankel family is included as an adjacent result; Q1160 concerns the **cycle** family.

How to read the certificates

An all-parameter theorem settles its displayed quantifiers. An exact polynomial root certificate settles that complete polynomial. A factorization of a finite matrix into nonnegative bidiagonal matrices settles every minor of that finite matrix. A passed range of unrelated instances remains finite evidence for the family.

This distinction matters especially for Q2400: positive expectations or favorable samples do not prove its high-probability limit. The partial random-graph theorem below supplies its own explicit probability bound. Likewise, the exact diagnostic never turns missing prefix terms into zeros and never uses leading principal minors alone as a semidefinite-matrix criterion.

Contents of the accompanying package

The consolidated report contains the mathematical arguments. The archive also contains the individual proof notes, a standalone Python diagnostic, exact certificate data, independent controls, and reconstruction scripts. Most scripts use only Python's standard library. The two additional symbolic audits use SymPy; the exhaustive atom screen uses C++20. The report's numerical root-locus illustration visualizes an analytic theorem and is not used as its proof.

A sharp imaginary-part bound for Stanley-nonnegative polynomials

Research derivation, 6 October 2026. Statement Q325, ID 4dda0904c7526b4a8b45.

Result and exact scope

For an integer $d \geq 1$, let

$$b_j(z) = \binom{z+d-j}{d}, \quad 0 \leq j \leq d,$$

and let S_d consist of the nonzero polynomials

$$f(z) = \sum_{j=0}^d a_j b_j(z), \quad a_j \geq 0.$$

Set $M_d = b_0 + b_d$ and let B_d be the largest absolute imaginary part of a zero of M_d .

Theorem. Every zero z of every polynomial in S_d satisfies

$$|\operatorname{Im} z| \leq B_d.$$

For $d \geq 2$, equality holds for at least one zero of f if and only if f is a positive scalar multiple of M_d . Every zero attaining equality has real part $-1/2$.

This is the exact general bound asked by Braun-Develin's Conjecture 2.4 and by Q325; it assumes no canonical-line condition on the other zeros.

A centered interval lemma

Fix $y > 0$ and $h > 0$. Define

$$J_{y,h}(c) = \int_{c-h}^{c+h} \frac{y}{t^2 + y^2} dt.$$

Then $J_{y,h}$ is even and strictly decreases as c increases from zero. Indeed,

$$J'_{y,h}(c) = \frac{y}{(c+h)^2 + y^2} - \frac{y}{(c-h)^2 + y^2},$$

which is negative for $c > 0$, because $(c+h)^2 - (c-h)^2 = 4ch > 0$. Consequently

$$J_{y,h}(c) \leq J_{y,h}(0) = 2 \arctan(h/y),$$

with equality precisely when $c=0$.

Proof of the theorem

When $d=1$, $f(z) = (a_0 + a_1)z + a_0$ has a real zero, and $M_1(z) = 2z+1$, so the result is immediate. Assume $d \geq 2$.

For $z = x + iy$, $y > 0$, define $\arg(z+r)$ to be the argument in $(0, \pi)$. None of the factors vanish. Define real-valued lifted phases

$$\Phi_j(z) = \sum_{k=1}^d \arg(z + k - j), \quad 0 \leq j \leq d.$$

These are arguments of $b_j(z)$ modulo 2π ; no reduction modulo 2π is made when comparing them. Cancellation of the common factors yields

$$\Phi_{j+1}(z) - \Phi_j(z) = \arg(z - j) - \arg(z + d - j) > 0. \quad (1)$$

The strict inequality follows because $\arg(t+iy)$ strictly decreases as t increases. Thus all the lifted phases lie in the interval $[\Phi_0, \Phi_d]$.

Write $\Theta_d(x, y) = \Phi_d - \Phi_0$. Reordering the terms in the two endpoint sums, without changing any individual argument, gives

$$\begin{aligned} \Theta_d(x, y) &= \sum_{k=1}^d [\arg(x + 1 - k + iy) - \arg(x + k + iy)] \\ &= \sum_{k=1}^d \int_{x+1-k}^{x+k} \frac{y}{t^2 + y^2} dt. \end{aligned} \quad (2)$$

Every integral in (2) has center $x+1/2$; the k th has half-width $k-1/2$. The centered interval lemma therefore proves

$$\Theta_d(x, y) \leq F_d(y) := 2 \sum_{k=1}^d \arctan\left(\frac{k-1/2}{y}\right), \quad (3)$$

with strict inequality unless $x = -1/2$.

The function $F_d(y)$ is continuous and strictly decreasing on $(0, \infty)$, with limits $d\pi$ as y tends to zero and 0 as y tends to infinity. Hence there is a unique $\beta_d > 0$ satisfying

$$F_d(\beta_d) = \pi. \quad (4)$$

If $y > \beta_d$, equations (1)-(4) show that all the lifted phases lie in an interval of length strictly less than π . Multiplication by $\exp(-i(\Phi_0 + \Phi_d)/2)$ therefore sends every nonzero complex number $b_j(z)$ into the open right half-plane. A nonzero nonnegative linear combination of these numbers has strictly positive real part after the same rotation, so it cannot vanish. Thus every f in S_d has no zeros with imaginary part greater than β_d .

At $z_* = -1/2 + i\beta_d$, equation (4) says that the endpoint phases differ by exactly π . Also

$$|b_0(z_*)| = |b_d(z_*)|,$$

because the magnitudes of their factors are, in opposite orders,

$$\sqrt{(k-1/2)^2 + \beta_d^2}, \quad 1 \leq k \leq d.$$

Consequently $b_0(z_*) + b_d(z_*) = 0$, so $M_d(z_*) = 0$. We have already proved that no member of S_d , including M_d , has a zero with imaginary part greater than β_d . Since all coefficients are real, conjugation handles the lower half-plane as well. It follows that $B_d = \beta_d$, proving the desired bound.

For equality, suppose $f(x+iB_d)=0$. If $x \neq -1/2$, the strict part of (3) still puts all terms in an open half-plane, which is impossible. Thus $x=-1/2$. Equations (1) and (4) put the phases of all interior terms, $1 \leq j \leq d-1$, strictly between the two endpoint phases. After rotation, the endpoint terms have real part zero and all interior terms have strictly positive real part. Therefore $a_j=0$ for all interior indices. The equal endpoint magnitudes and opposite endpoint directions then force $a_0=a_d>0$. The converse follows from $M_d(z_*)=0$. This proves the equality assertion.

Stronger result: the complete nonreal root locus

For $y>0$, let $\Theta_d(x, y)$ be as in (2). Then

$$x + iy \text{ is a root of some } f \in S_d \iff \Theta_d(x, y) \geq \pi. \quad (5)$$

The forward direction is already proved by the open-half-plane argument. For the converse, suppose $\Theta_d \geq \pi$. Every successive phase increment in (1) is strictly between 0 and π . Let m be the smallest index such that $\Phi_m - \Phi_0 \geq \pi$. Set

$$\beta = \Phi_{m-1} - \Phi_0, \quad \gamma = \Phi_m - \Phi_0.$$

Minimality and the strict increment bounds give

$$0 < \beta < \pi \leq \gamma < 2\pi, \quad 0 < \gamma - \beta < \pi.$$

The identity

$$\sin(\gamma - \beta) - \sin \gamma e^{i\beta} + \sin \beta e^{i\gamma} = 0 \quad (6)$$

now supplies an explicit SNN polynomial vanishing at the specified point. Take

$$a_0 = \frac{\sin(\gamma - \beta)}{|b_0(z)|}, \quad a_{m-1} = \frac{-\sin \gamma}{|b_{m-1}(z)|}, \quad a_m = \frac{\sin \beta}{|b_m(z)|},$$

and take every other coefficient to be zero. The first and last of these coefficients are strictly positive, and the middle one is nonnegative. Multiplying (6) by $\exp(i \Phi_0)$ proves $\sum a_j b_j(z) = 0$. The indices are distinct because $m \geq 2$.

Thus every nonreal point in the root locus is realized by at most three binomial-basis terms. When the initial crossing occurs exactly at $\gamma=\pi$, two suffice. This converse and its explicit witness were obtained after independent discussion with the coordinating agent.

For a computational formula without complex arguments, (2) reads

$$\sum_{k=1}^d \left[\arctan\left(\frac{x+k}{y}\right) - \arctan\left(\frac{x+1-k}{y}\right) \right] = \Theta_d(x, y). \quad (7)$$

Formula (7) uses the ordinary real arctangent in $(-\pi/2, \pi/2)$ and therefore has no branch ambiguity. Complex conjugation describes the lower-half-plane locus.

For completeness, the real root locus is precisely $[-d, d-1]$. Outside this interval all binomial basis polynomials have the same strict sign. At an integer in the interval at least one basis polynomial vanishes. At a noninteger x in the interval one can choose $0 \leq j \leq d-1$ with $j-d < x < j$; then $(x+d-j)/(x-j) < 0$, so $b_j(x)$ and $b_{j+1}(x)$ have opposite signs. A positive combination of this pair vanishes at x .

Low-degree controls

- $M_1(z)=2z+1$, $B_1=0$.
- $M_2(z)=z^2+z+1$, $B_2=\sqrt{3}/2$.
- $M_3(z)=(2z+1)(z^2+z+6)/6$, $B_3=\sqrt{23}/2$.

The companion script `verify_interpolation.py` computes integer multiples $d!$ M_d exactly, checks the endpoint-phase identities numerically as a secondary diagnostic, and verifies these exact low-degree formulas. The proof does not depend on these computations.

Literature verification and novelty qualification

1. Benjamin Braun and Mike Develin, Ehrhart Polynomial Roots and Stanley’s Non-negativity Theorem, arXiv:math/0610399 (2006), [primary PDF](#). Definition 1.2 defines SNN polynomials; Conjecture 2.4 is the exact imaginary-part bound above. Their equation (2.2) identifies the same value B_d by the arctangent sum in (4). Their discussion notes the degree-two case.
2. Max Kölbl, Properties of Ehrhart Polynomials Whose Roots Lie on the Canonical Line, *Integers* 25 (2025), A68, [primary PDF](#). Conjecture 2 retains the general bound; Section 2 proves it under the additional canonical-line hypothesis.

The derivation above was obtained in this research session. A focused search of these exact conjecture phrases found no prior general proof. That is a qualified prior-art check, not a guarantee of novelty. The mathematical statement and proof should receive further independent review before any durable external resolution or publication claim.

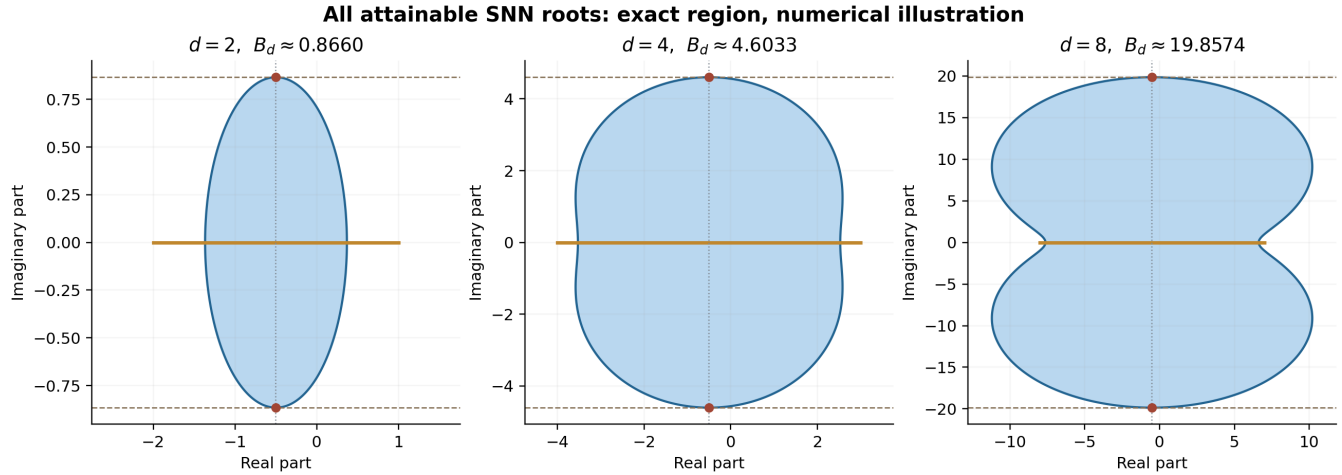


Figure 1: Blue regions are the attainable nonreal roots; the gold segment gives all attainable real roots. The dashed lines are the proved sharp bounds. The red points have real part $-1/2$ and imaginary parts $\pm B_d$. Numerical contour sampling is illustrative only.

Q165: an all-length coefficient certificate and a Hurwitz-stability proof

Date: 2026-10-06. Status: complete proof supplied below, with exact symbolic verification in `verify_q165_certificate.py` and independent audits of the formal-series, algebraic, and analytic steps. This establishes the stated parity real-rootedness conjecture through the stronger conclusion that every nonzero zero of the path domination polynomial lies in the open left half-plane. No claim of literature priority is made.

Statement and the key positive polynomial

Let

$$D_0(x) = 1, \quad D_1(x) = x, \quad D_2(x) = x^2 + 2x, \quad D_n(x) = x(D_{n-1}(x) + D_{n-2}(x) + D_{n-3}(x)) \quad (n \geq 3).$$

These are the domination polynomials of paths. The zero at the origin has multiplicity $\lceil n/3 \rceil$: positivity of the recurrence coefficients gives $m_n = 1 + \min(m_{n-1}, m_{n-2}, m_{n-3})$ and the stated formula follows from the three initial values. For real y , define

$$F_n(y^2) = \operatorname{Re}(D_n(iy)D_{n-1}(-iy)).$$

This defines an integer polynomial $F_n(z)$. We prove the following stronger coefficient statement:

Theorem 1. Every coefficient of every $F_n(z)$ is nonnegative. Moreover $F_n(z) > 0$ for $n \geq 2$ and real $z > 0$.

The first values are

$$F_1 = 0, \quad F_2 = 2z, \quad F_3 = 2z + z^2, \quad F_4 = 8z^2 + z^3, \quad F_5 = 12z^2 + 9z^3 + z^4.$$

An exact rational generating function

Put $G(t, z) = \sum_{n \geq 1} F_n(z)t^n$. Then

$$G(t, z) = \frac{N(t, z)}{D(t, z)}, \tag{1}$$

where

$$\begin{aligned} D(t, z) = & 1 - z(t + 2t^2 + 3t^3) \\ & + z^2(t^3 - 4t^4 + t^5 + 3t^6) \\ & - z^3(t^5 - t^6 - t^8 + t^9), \end{aligned} \tag{2}$$

and

$$\begin{aligned} N(t, z) = & t^2 z (2 + 2t - tz + 2t^2 z + 2t^3 z - 3t^4 z \\ & + t^3 z^2 - t^4 z^2 + t^5 z^2 - 2t^6 z^2 + t^7 z^2). \end{aligned} \tag{3}$$

Here is a finite exact derivation of this identity. Set

$$T(x) = \begin{pmatrix} x & x & x \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad K = T(x) \otimes T(-x).$$

For $n \geq 2$, the product vector of $(D_n(x), D_{n-1}(x), D_{n-2}(x))$ and its evaluation at $-x$ evolves by K . Its symmetric $(0, 1), (1, 0)$ component is $F_n(-x^2)$. Direct determinant expansion gives

$$\begin{aligned} \det(qI - K) &= q^9 + x^2q^8 + 2x^2q^7 + (x^4 + 3x^2)q^6 - 4x^4q^5 \\ &\quad + (x^6 + x^4)q^4 + (-x^6 + 3x^4)q^3 - x^6q + x^6. \end{aligned}$$

Reversing this characteristic polynomial and substituting $x^2 = -z$ gives (2). Cayley-Hamilton makes its recurrence valid for F_n at $n \geq 11$. Multiplying the initial series through F_{10} by (2) gives exactly (3), with zero coefficient of t^{10} . This proves (1); it is not a guessed recurrence from finite data. The verification script reconstructs both the determinant and the initial multiplication independently.

Change variables to expose a positive recurrence

Use the monomial change

$$w = zt^3, \quad u = t^{-1}.$$

Write $H(w, u) = \tilde{N}(w, u)/\tilde{D}(w, u)$ for the rational function obtained from (1). The numerator and denominator become

$$\begin{aligned} \tilde{D} &= (1 - w)^3 - u w(1 - w)(w + 2) - u^2 w(1 + 4w) \\ &\quad + u^3 w^2(1 + w) - u^4 w^3, \end{aligned} \tag{4}$$

$$\begin{aligned} \tilde{N} &= w(1 - w)(2 - w) + 2u w(1 + w - w^2) + u^2 w^2(2 + w) \\ &\quad - u^3 w^2(1 + w) + u^4 w^3. \end{aligned} \tag{5}$$

The occurrence of t^{-1} causes no formal-series ambiguity. Every monomial $w^j u^r$ in $\tilde{N}$ and in $\tilde{D} - 1$ satisfies $j \geq 1$ and $0 \leq r \leq 2j$. This cone is closed under multiplication, so the geometric inverse $1/\tilde{D}$, and hence H , are supported in the same cone (with a possible constant for the inverse). Mapping such a monomial back gives $z^j t^{3j-r}$, whose t -degree is at least j . There are only finitely many contributing pairs for any fixed t -degree. Consequently this substitution defines a legitimate series map, and the resulting series is (1) by uniqueness of the inverse of $D(t, z)$. It will therefore suffice to prove that every coefficient of $H(w, u)$ is nonnegative.

Normalize (4) by $(1 - w)^3$, and multiply both numerator and denominator by

$$1 + \rho u, \quad \rho = \frac{w}{1 + w}.$$

Although ρ itself has alternating coefficients, the resulting denominator is

$$1 - \alpha u - \beta u^2 - \gamma u^3 - \eta u^5, \tag{6}$$

where

$$\begin{aligned}
\alpha &= \frac{w(1+5w)}{(1-w)(1-w^2)}, \\
\beta &= \frac{w(1+4w)}{(1-w)^3} + \frac{w^2(w+2)}{(1-w)(1-w^2)}, \\
\gamma &= \frac{w^3(2-w)}{(1-w)^2(1-w^2)}, \\
\eta &= \frac{w^4}{(1-w)^2(1-w^2)}.
\end{aligned} \tag{7}$$

All four are nonnegative coefficientwise as power series in w . The only slightly nonobvious case follows from

$$\frac{2-w}{(1-w)^2} = \sum_{j \geq 0} (j+2)w^j.$$

Let the transformed numerator be

$$M(w, u) = \frac{\tilde{N}(w, u)(1 + \rho u)}{(1-w)^3}.$$

It has degree at most five in u , and

$$[u^4]M = 0, \quad [u^5]M = \eta. \tag{8}$$

Indeed, before division by $(1-w)^3$, its u^4 coefficient is $w^3 - \rho w^2(1+w) = 0$, and its u^5 coefficient is ρw^3 .

The four initial series and positivity

Expand $H(w, u) = \sum_{r \geq 0} g_r(w)u^r$. Division of (5) by (4) gives

$$\begin{aligned}
g_0 &= \frac{w(2-w)}{(1-w)^2}, \\
g_1 &= \frac{w(w^3 - 4w^2 + 4w + 2)}{(1-w)^4}, \\
g_2 &= \frac{w^2(3w^3 - 12w^2 + 10w + 8)}{(1-w)^6}, \\
g_3 &= \frac{w^2(1 + 28w + 28w^2 - 43w^3 + 14w^4 - w^5)}{(1-w)^8}.
\end{aligned} \tag{9}$$

The signed numerators do not obstruct coefficient positivity. Their coefficients are explicitly as follows. For $j \geq 1$,

$$[w^j]g_0 = j+1, \quad [w^j]g_1 = \frac{(j+1)(j^2 + 3j - 2)}{2}. \tag{10}$$

Their constant coefficients are zero. For $m \geq 0$,

$$[w^{m+2}]g_2 = \frac{(m+1)(m+2)(9m^3 + 133m^2 + 538m + 480)}{120}, \quad (11)$$

$$[w^{m+2}]g_3 = \frac{(m+1)(m+2)(9m^5 + 211m^4 + 1575m^3 + 4163m^2 + 3282m + 840)}{1680}. \quad (12)$$

The coefficients of degrees zero and one in g_2, g_3 are zero. These formulas follow directly by expanding $(1-w)^{-k}$, and are exact polynomial identities. In particular all four initial series are nonnegative coefficientwise.

Equations (6)–(8), with $g_r = 0$ for $r < 0$, now imply

$$g_4 = \alpha g_3 + \beta g_2 + \gamma g_1,$$

$$g_5 = \alpha g_4 + \beta g_3 + \gamma g_2 + \eta g_0 + \eta,$$

and, for every $r \geq 6$,

$$g_r = \alpha g_{r-1} + \beta g_{r-2} + \gamma g_{r-3} + \eta g_{r-5}. \quad (13)$$

Induction proves that every g_r is nonnegative coefficientwise. The justified series substitution in Section 3 proves the nonnegativity in Theorem 1. Finally $F_2 = 2z$, while $[z^{n-1}]F_n = 1$ for $n \geq 3$: the latter follows by using the top two terms $D_n(x) = x^n + nx^{n-1} + \dots$ and $D_{n-1}(x) = x^{n-1} + (n-1)x^{n-2} + \dots$. Thus $F_n(z) > 0$ for $n \geq 2, z > 0$. This completes the proof of Theorem 1.

From the coefficient certificate to Hurwitz stability

Theorem 2. For every $n \geq 1$, all nonzero zeros of D_n have strictly negative real part.

The cases $n = 1, 2$ are immediate. Assume the conclusion holds for D_{n-1} and let

$$R_n(x) = \frac{D_n(x)}{D_{n-1}(x)}.$$

There are no poles in the closed right half-plane. At the origin the quotient has a removable singularity because the zero multiplicities $\lceil n/3 \rceil$ are nondecreasing. Its removable value there is non-negative: either zero, or a ratio of positive lowest coefficients. At all other points of the imaginary axis,

$$\operatorname{Re} R_n(iy) = \frac{F_n(y^2)}{|D_{n-1}(iy)|^2} > 0.$$

For $n \geq 3$, polynomial division at infinity gives $R_n(x) = x + 1 + O(1/x)$. On a sufficiently large right semicircle, therefore, $\operatorname{Re} R_n(x) > 0$. The minimum principle for the harmonic function $\operatorname{Re} R_n$, applied on the corresponding right half-disk, gives nonnegativity in its interior; the strong minimum principle gives strict positivity there, since this harmonic function is not constant. Taking arbitrarily large radii proves strict positivity throughout the open right half-plane. The quotient consequently has no zeros there; its strictly positive real part on the nonzero imaginary axis also excludes such boundary zeros. Induction proves Theorem 2.

Resolution of Q165

Write $D_n(x) = x^m P_n(x)$, with $m = \lceil n/3 \rceil$. By Theorem 2 the polynomial P_n is strictly Hurwitz stable and has positive coefficients. The classical Hermite–Biehler theorem implies that its even and odd component polynomials have only negative real roots, with the usual harmless constant or zero component in the smallest cases.

Restoring the factor x^m only adds zero roots and possibly interchanges the two components. Explicitly, if $P_n(x) = E(x^2) + xO(x^2)$, then for $m = 2a$ the parity components of D_n are $x^a E(x)$ and $x^a O(x)$; for $m = 2a + 1$ they are $x^{a+1} O(x)$ and $x^a E(x)$. Therefore both polynomials in Q165 have only real nonpositive roots. The identically zero component of $D_1 = x$ is interpreted in the conventional degenerate sense; every nonzero parity component satisfies the assertion literally.

Exact verification and sources

`verify_q165_certificate.py` verifies the transfer characteristic polynomial, the generating-function numerator, the transformed denominator, the two numerator tail identities, the four initial rational functions, and the coefficient formulas by exact symbolic algebra. It also checks the rational generating-function identity against independently generated F_n through $n = 40$. These finite checks audit the displayed proof identities; the all-length conclusion follows from Cayley–Hamilton and the positive recurrence (13).

Primary sources used for background, not for the new displayed certificate:

- Yufei Zhang, Some Combinatorial Problems Involving Total Non-Negativity and Unimodality, printed pp. 82–83, Conjecture 5.1.4: <https://ndownloader.figshare.com/files/54098216> .
- Galvin–Zhang, arXiv:2408.12731, path domination recurrence: <https://arxiv.org/html/2408.12731v1> .
- Brown–Cameron, On the Stability of Independence Polynomials, arXiv:1802.02478, Theorem 2.1 (PDF p. 4), for the even/odd form of Hermite–Biehler: <https://arxiv.org/pdf/1802.02478> .

This proof supersedes the earlier unresolved-status note `q165-hurwitz-route.md`, whose bounded checks remain valid provenance.

Q166: A four-functions proof of log-concavity

Date: 2026-10-06. Status: structural proof, independently audited by the root research agent; prior-art status unclaimed. The proof establishes the exact conjecture stated in the handoff, for every positive integer n and every integer $\ell \geq 2$. It also establishes a weighted strengthening and ordinary log-concavity for all powers of paths (a weaker property than the ULC requested in Q164).

General interval-constraint theorem

Let $m \geq 0$. Fix any collection of integer bounds

$$L_{ab} \leq \sum_{i=a+1}^b u_i \leq U_{ab}, \quad 0 \leq a < b \leq m,$$

where u is a binary word of length m . Either bound may be absent. Let $\mathcal{F}$ be the set of words satisfying all bounds, and write

$$f_t = \# \left\{ u \in \mathcal{F} : \sum_i u_i = t \right\},$$

with $f_t = 0$ outside $[0, m]$. Then

$$f_t^2 \geq f_{t-1} f_{t+1}$$

for every integer t . More generally, for arbitrary nonnegative weights $w_1, \dots, w_m$,

$$f_t(w) = \sum_{\substack{u \in \mathcal{F} \\ \sum_i u_i = t}} \prod_{i=1}^m w_i^{u_i}$$

is log-concave in t . This is a numerical statement for all nonnegative choices of the weights; no claim of coefficientwise log-concavity is made here.

Proof, unweighted case

Represent u by its prefix-height vector $h = (h_0, \dots, h_m)$, where $h_0 = 0$ and $h_j = \sum_{i=1}^j u_i$. The binary restrictions become

$$h_{i-1} \leq h_i \leq h_{i-1} + 1.$$

Every interval restriction is a pair of difference inequalities $h_b \leq h_a + U_{ab}$ and $h_a \leq h_b - L_{ab}$.

Let K be the set of all integer vectors satisfying these difference inequalities, with h_0 free. The set K is invariant under uniform integer translation, $h \mapsto h + c\mathbf{1}$. It is also closed under coordinatewise minimum and maximum. Indeed, if $x_b \leq x_a + c$ and $y_b \leq y_a + c$, monotonicity of minimum gives

$$\min(x_b, y_b) \leq \min(x_a + c, y_a + c) = \min(x_a, y_a) + c,$$

and the identical argument with maximum proves closure under maximum. Apply this to every difference inequality.

Let

$$H_t = \{h \in K : h_0 = 0, h_m = t\}.$$

These are the prefix-height versions of the rank- t words, so $|H_t| = f_t$. Fix $1 \leq t \leq m-1$ and form the finite sets

$$A = H_{t-1} + \mathbf{1}, \quad B = H_{t+1}.$$

For every $a \in A$ and $b \in B$, coordinatewise meet and join are in K . Their endpoint values are

$$\begin{aligned} (a \wedge b)_0 &= 0, & (a \wedge b)_m &= t, \\ (a \vee b)_0 &= 1, & (a \vee b)_m &= t+1. \end{aligned}$$

Consequently,

$$A \wedge B \subseteq H_t, \quad A \vee B \subseteq H_t + \mathbf{1}.$$

All four sets lie in the finite product lattice $\{0, \dots, m+1\}^{m+1}$. The cardinality case of the Ahlswede-Daykin four-functions theorem gives

$$|A| |B| \leq |A \wedge B| |A \vee B| \leq |H_t|^2,$$

which is the required inequality. Boundary inequalities are automatic because the out-of-range terms are zero. This argument does not assert that taking meet and join is an injection on ordered pairs; the four-functions theorem supplies precisely the counting inequality that replaces that invalid assertion.

Weighted extension

First suppose every w_i is positive, and on the finite product lattice define

$$W(h) = \prod_{i=1}^m w_i^{h_i - h_{i-1}}.$$

The weight W is invariant under uniform translation, and

$$W(a)W(b) = W(a \wedge b)W(a \vee b),$$

because coordinatewise $a + b = (a \wedge b) + (a \vee b)$. Apply the weighted four-functions theorem to the four functions W times the indicators of A , B , H_t , and $H_t + \mathbf{1}$. Their pointwise product inequality follows from the displayed identity and the two set inclusions above. Summation gives

$$f_{t-1}(w)f_{t+1}(w) \leq f_t(w)^2.$$

Taking limits of positive weights at any zero-weight positions proves the nonnegative case by continuity.

Application to forbidden runs of one-parts

A composition of n into k positive parts corresponds uniquely to a set of $k-1$ cuts among the $n-1$ internal positions $1, \dots, n-1$. Write $c_i = 1$ if there is a cut at i and $c_i = 0$ otherwise, and adjoin the two fixed boundary cuts $c_0 = c_n = 1$.

There are ℓ consecutive parts equal to 1 exactly when there is some i with $0 \leq i \leq n-\ell$ for which

$$c_i = c_{i+1} = \dots = c_{i+\ell} = 1.$$

There are $\ell + 1$ cuts in this display because ℓ unit parts have $\ell + 1$ endpoints. Thus the avoidance condition is exactly

$$\sum_{j=i}^{i+\ell} c_j \leq \ell \quad \text{for every } 0 \leq i \leq n - \ell.$$

For each such interval subtract the contribution of the fixed boundary cuts. The resulting constraint is an upper bound on the sum of the internal cut indicators in an interval of $\{1, \dots, n-1\}$. The general theorem applies with $m = n - 1$ and $t = k - 1$, proving

$$A_\ell(n, k)^2 \geq A_\ell(n, k-1)A_\ell(n, k+1).$$

If $n < \ell$ there are no avoidance constraints and the usual binomial composition sequence is recovered. If $n = \ell$, the window from 0 to n forbids precisely the all-ones composition. If $n = 1$, the single composition (1) is allowed because $\ell \geq 2$, and the one-term sequence is log-concave. For $n = k$, the all-ones composition is allowed if and only if $n < \ell$, as required. No exceptional boundary formula is imported from the thesis.

The weighted extension allows an arbitrary nonnegative multiplicative weight for each internal cut position. It does not require a common ℓ : one can forbid independently specified runs located in specified intervals, or impose other upper or lower bounds on cut counts in intervals.

Application to Q164: ordinary log-concavity for path powers

A dominating set D in P_n^ℓ is represented by $d_i = \mathbf{1}_{\{i \in D\}}$. It is dominating exactly when

$$\sum_{j=\max(1, i-\ell)}^{\min(n, i+\ell)} d_j \geq 1 \quad \text{for every } i = 1, \dots, n.$$

These are interval lower bounds. The general theorem therefore proves that the domination coefficient sequence of P_n^ℓ is log-concave for all n and ℓ , and remains log-concave when every vertex receives an arbitrary fixed nonnegative multiplicative weight.

The same argument applies to the dominating sets of any graph equipped with a vertex ordering in which each closed neighborhood is an interval, or more generally to hitting sets of any interval hypergraph. It does not by itself prove ultra-log-concavity after division by $\binom{n}{k}$, and the circular-neighborhood constraints for a cycle are not all single intervals in one linear order. Accordingly, Q164 remains open in its stated full strength in this report.

Sources and verification

- Yufei Zhang, Some Combinatorial Problems Involving Total Non-Negativity and Unimodality, Notre Dame dissertation, 2025. Conjecture 5.1.7 on printed p. 86 is Q166. Definition/sequence convention for Q167 is on pp. 91-92. <https://ndownloader.figshare.com/files/54098216> ; institutional landing page https://curate.nd.edu/articles/thesis/Some_Combinatorial_Problems_Involving_Total_Non-Negativity_and_Unimodality/28786115 .
- Rudolf Ahlswede and David E. Daykin, An Inequality for the Weights of Two Families of Sets, Their Unions and Intersections, Z. Wahrscheinlichkeitstheorie verw. Gebiete 43 (1978), 183-185. Theorem 1 and the immediate distributive-lattice corollary on the first page. Primary author-hosted PDF: <https://www.math.uni-bielefeld.de/ahlsweide/homepage/public/30.pdf> .

- David Galvin and Yufei Zhang, The domination polynomial of powers of paths and cycles, arXiv:2408.12731, 2024. <https://arxiv.org/html/2408.12731v1> . The paper establishes unimodality and states the ULC conjecture; the present proof strengthens its path portion to ordinary log-concavity but does not prove that full conjecture.
- Mathlib’s independently formalized four-functions theorem and cardinality corollary document the exact form used above: https://leanprover-community.github.io/mathlib4_docs/Mathlib/Combinatorics/SetFamily/FourFunctions.html . This is corroboration, not the primary mathematical source.

The decisive result is the proof. Finite checks are separate implementation controls and are recorded in `q166-controls.json` after execution of `verify_q166.py`.

Higher-order Stirling positivity: structural results and exact certificates

Date: 2026-10-06.

Results and scope

The four assigned B3 targets retain their full infinite conjectures. The following all-parameter results substantially restrict what a counterexample could look like.

Target	Structural result established here	Remaining question
Q1158: total nonnegativity of the cycle triangle	Every minor of order at most two is nonnegative, for every integer $r \geq 1$.	Minors of order at least three.
Q1159: total nonnegativity of the subset triangle	Every minor of order at most two is nonnegative, for every integer $r \geq 1$.	Minors of order at least three.
Q1160: coefficientwise Hankel positivity of cycle rows	For every $r \geq 2$ and every Hankel minor, its highest-degree coefficient and its lowest nonzero coefficient are strictly positive. Their degrees are explicit.	The coefficients strictly between those two degrees.
Q1161: third-order left-half-plane stability	Exact Routh certificates for both families, every n from 1 through 80. After the simple zero at zero is removed, every remaining root lies in the open left half-plane.	An all- n proof.

There is also an all-parameter negative certificate for the **related subset Hankel question**, Deb-Sokal Conjecture 1.4(d): a single specified 3×3 minor has a negative coefficient for every $r \geq 3$. This is adjacent to Q1160 but is not one of the twenty exact B3 statement IDs. It must not be reported as a disproof of Q1160.

The proofs below are mathematical certificates. The bounded calculations later in this report certify their stated finite instances and do not extend their bounds by inference.

Sources and status qualification

The definitions and recurrence are supplied in the user attachment and in [Deb-Sokal, arXiv:2507.18959](#), Lemma 1.1. The source identifies Q1158-Q1160 as parts of Conjectures 1.3-1.4 and Q1161 as Conjectures 7.5 and 7.7. Appendix C already reports triangle tests through 70×70 for $r = 2, \dots, 10$; cycle Hankel tests reach 11×11 for $r = 3, \dots, 7$ and 10×10 for $r = 8, 9, 10$. Section 7 uses numerical zero plots at $n = 50, 100, 150, 200$. No later resolution of these exact four targets was located in the focused searches performed for this report; this is not an exhaustive literature certification.

A manuscript dated 2 September 2026 by Carptopus already states log-concavity of the normalized cycle rows used below. The short proof needed here is independently reproduced in full. Its much longer claims about other log-concavity cases are not relied upon. No claim of first publication is made for the present consequences.

Definitions

Write $p = r-1$. For A equal to C or S , let $A(n,k)$ denote the higher-order, column-shifted triangle from the handoff. Thus $A(0,0)=1$; other entries outside $0 \leq k \leq n$ vanish; and, for $n \geq 1$,

$$C(n, k) = (n + pk - 1)^p C(n-1, k-1) + (n + pk - 1) C(n-1, k),$$

$$S(n, k) = \frac{(n + pk - 1)^p}{p!} S(n-1, k-1) + k S(n-1, k).$$

The falling factorial of order zero equals one. Entries with $1 \leq k \leq n$ are strictly positive. Define

$$c_{r,n}(x) = \sum_k C(n, k) x^k, \quad s_{r,n}(x) = \sum_k S(n, k) x^k.$$

All matrix row and column indices below are these shifted indices, not the original sizes of the underlying permutations or partitions.

Every higher-order Stirling triangle is TN of order two

2.1 An elementary preservation lemma

Let (x_k) be a nonnegative log-concave sequence with no internal zeros, extended by zero outside its support. Suppose c_k, d_k are nonnegative, log-concave on the indices needed, and satisfy

$$2c_k d_k \geq c_{k-1} d_{k+1} + c_{k+1} d_{k-1}. \quad (2.1)$$

Then $y_k = d_k x_k + c_k x_{k-1}$ is log-concave.

Proof. Expansion gives

$$\begin{aligned} y_k^2 - y_{k-1} y_{k+1} &= d_k^2 x_k^2 - d_{k-1} d_{k+1} x_{k-1} x_{k+1} \\ &\quad + c_k^2 x_{k-1}^2 - c_{k-1} c_{k+1} x_{k-2} x_k \\ &\quad + (2c_k d_k - d_{k-1} c_{k+1}) x_{k-1} x_k - c_{k-1} d_{k+1} x_{k-2} x_{k+1}. \end{aligned}$$

Log-concavity implies $x_k^2 \geq x_{k-1} x_{k+1}$, $x_{k-1}^2 \geq x_{k-2} x_k$, and $x_{k-1} x_k \geq x_{k-2} x_{k+1}$. The last inequality follows by chaining the decreasing adjacent ratios on the positive support; it also holds at the zero boundaries. Consequently the displayed difference is at least

$$(d_k^2 - d_{k-1} d_{k+1}) x_k^2 + (c_k^2 - c_{k-1} c_{k+1}) x_{k-1}^2 + (2c_k d_k - d_{k-1} c_{k+1} - c_{k-1} d_{k+1}) x_{k-1} x_k,$$

which is nonnegative. $\square$

This is the elementary log-concavity criterion commonly attributed to Sagan; the proof above supplies all that is used.

2.2 Normalized rows are log-concave for every r

For either triangle define

$$H(n, k) = \frac{k! A(n, k)}{(n + pk)!}. \quad (2.2)$$

For a fixed recurrence step n , put $N = n + pk$. Direct factorial cancellation gives

$$H_C(n, k) = \frac{N-1}{N} H_C(n-1, k) + \frac{k}{N} H_C(n-1, k-1), \quad (2.3)$$

$$H_S(n, k) = \frac{k}{N} \left(H_S(n-1, k) + \frac{1}{p!} H_S(n-1, k-1) \right). \quad (2.4)$$

To apply the lemma to (2.3), use $c_k = k/N$ and $d_k = (N-1)/N$. For $n \geq 2$ and $1 \leq k \leq n-1$, set

$$Q = N^2(N-p)(N+p) > 0.$$

Elementary common-denominator calculations give

$$\begin{aligned} d_k^2 - d_{k-1}d_{k+1} &= \frac{p^2(2N-1)}{Q}, \\ c_k^2 - c_{k-1}c_{k+1} &= \frac{n(2N-n)}{Q}, \\ 2c_kd_k - c_{k-1}d_{k+1} - c_{k+1}d_{k-1} &= \frac{2p((n+1)N-n)}{Q}. \end{aligned} \quad (2.5)$$

Every numerator is nonnegative for $p \geq 0$. At $p=0$ the first and third expressions vanish, which causes no problem. The $n=0$ and $n=1$ rows and the zero boundaries are immediate. Induction proves log-concavity of all normalized cycle rows.

For (2.4), the two coefficient sequences are proportional to $k/(n+pk)$, whose log-concavity is the second identity above. Their mixed defect is a positive constant times that same log-concavity defect. The lemma therefore proves log-concavity of all normalized subset rows as well. Equivalently, (2.4) first convolves the previous row with the positive two-term sequence $(1, 1/p!)$ and then multiplies termwise by a log-concave sequence.

2.3 The normalization supplies exactly the weight needed for TN2

Write

$$A(n, k) = W(n, k)H(n, k), \quad W(n, k) = \frac{(n+pk)!}{k!}.$$

Let $u_{n,k}$ and $v_{n,k}$ be the coefficients of $A(n-1, k-1)$ and $A(n-1, k)$, respectively, in the original recurrence. For $k \geq 1$,

$$\frac{W(n, k)^2}{W(n, k-1)W(n, k+1)} = \frac{k+1}{k} \frac{N!^2}{(N-p)!(N+p)!} = \frac{k+1}{k} \frac{u_{n+1,k}}{u_{n+1,k+1}}. \quad (2.6)$$

The constant factor $1/p!$ in the subset u 's cancels. The log-concavity just proved therefore yields

$$u_{n+1,k+1} A(n, k)^2 \geq \frac{k+1}{k} u_{n+1,k} A(n, k-1) A(n, k+1). \quad (2.7)$$

For k at the last nonzero position, the right side is zero and the needed inequality holds directly. Put $a_k = A(n, k)$. The adjacent-row, adjacent-column minor is exactly

$$\begin{aligned}
D_{n,k} &= A(n, k)A(n+1, k+1) - A(n, k+1)A(n+1, k) \\
&= u_{n+1,k+1}a_k^2 - u_{n+1,k}a_{k-1}a_{k+1} + (v_{n+1,k+1} - v_{n+1,k})a_ka_{k+1}.
\end{aligned} \tag{2.8}$$

The difference $v_{n+1,k+1} - v_{n+1,k}$ is p for cycles and 1 for subsets. Equations (2.7)-(2.8) prove $D_{n,k} \geq 0$ for all relevant $n, k \geq 1$.

For completeness, local minors imply arbitrary 2×2 minors here as follows. On the strictly positive support, (2.8) says that $A(n+1, k)/A(n, k)$ is nondecreasing in k . Chaining across columns gives

$$A(n, k)A(n+1, l) \geq A(n, l)A(n+1, k), \quad k < l \leq n.$$

Thus $A(n, l)/A(n, k)$ is nondecreasing in $n \geq l$. Chaining across rows proves the desired inequality for every pair of nonconsecutive rows as well. If an earlier row lies below the smaller selected column, its selected entries are both zero. If it lies between the selected columns, the upper-right selected entry is zero and the determinant is a product of nonnegative entries. Column zero has only $A(0, 0) = 1$; minors involving it are either zero or reduce to a nonnegative entry. These cases cover the entire infinite triangle.

Theorem 2.1. For every integer $r \geq 1$, both $C^{(r)}$ and $S^{(r)}$ are totally nonnegative of order two. $\square$

This theorem is strictly weaker than full total nonnegativity. It rules out all order-one and order-two counterexamples to Q1158 or Q1159.

Extreme coefficients of every cycle Hankel minor are positive

Let $I = (i_1 < \dots < i_m)$ and $J = (j_1 < \dots < j_m)$ be finite increasing lists of nonnegative integers, $m \geq 1$. Put

$$\begin{aligned}
F_{I,J}^{(r)}(x) &= \det(c_{r, i_a + j_b}(x))_{a,b=1}^m, \\
D &= \sum_a i_a + \sum_b j_b, \quad \nu = m - \mathbf{1}_{\{0 \in I \cap J\}}.
\end{aligned}$$

Theorem 3.1. For every $r \geq 2$,

$$\deg F_{I,J}^{(r)} = D, \quad \text{ord}_{x=0} F_{I,J}^{(r)} = \nu,$$

and both coefficients $[x^D]F$ and $[x^\nu]F$ are strictly positive. This includes the 1×1 minor $c_{r,0} = 1$, for which $D = \nu = 0$.

3.1 A moment-determinant fact

If a positive measure μ on $(0, \infty)$ has all needed moments and has a positive density, then every generalized Hankel determinant

$$\det \left(\int t^{i_a + j_b} d\mu(t) \right)_{a,b=1}^m$$

is strictly positive. Indeed, expansion and Fubini give the identity

$$\frac{1}{m!} \int \det(t_b^{i_a})_{a,b=1}^m \det(t_b^{j_a})_{a,b=1}^m \prod_{b=1}^m d\mu(t_b). \tag{3.1}$$

On $0 < t_1 < \dots < t_m$, each generalized Vandermonde determinant is positive. One way to see this is to factor out the ordinary Vandermonde; the quotient is a Schur polynomial with nonnegative

integer coefficients, multiplied if needed by a power of t_b . The product of the two determinants is invariant under permutations of the t_b and is positive off the diagonals. Integrability follows from the moment assumptions, and a region with distinct positive t_b has positive measure. This proves strict positivity.

3.2 The highest coefficient

The diagonal counts give

$$d_n = [x^n]c_{r,n}(x) = \frac{(rn)!}{r^n n!} = r^{(r-1)n} \prod_{q=1}^{r-1} (q/r)_n. \quad (3.2)$$

The final equality can be checked without invoking the multiplication formula: both sides equal one at $n=0$ and have the same ratio at consecutive n . Let G_q be independent gamma random variables of shape q/r and scale one. Then

$$X = r^{r-1} \prod_{q=1}^{r-1} G_q \implies \mathbb{E}X^n = d_n. \quad (3.3)$$

The product has a positive density on $(0, \infty)$, and all its positive integer moments are finite. In every permutation term of F , the maximum possible degree is D . Consequently

$$[x^D]F_{I,J}^{(r)} = \det(d_{i_a+j_b}) > 0$$

by (3.1). No cancellation can lower the degree.

3.3 The lowest coefficient

For $n \geq 1$, the coefficient of x counts a single cycle and is

$$[x]c_{r,n}(x) = (n+r-2)! = \Gamma(n+r-1). \quad (3.4)$$

These are the moments of the unnormalized positive gamma measure $t^{\{r-2\}e} \{-t\}dt$. If 0 is not in both I and J , every entry of the selected Hankel matrix is divisible by x , and

$$[x^m]F = \det(\Gamma(i_a + j_b + r - 1)) > 0.$$

If 0 is in both I and J , the upper-left entry is $c_{\{r,0\}}=1$. A determinant term using that entry can have degree $m-1$; every term not using it has degree at least m . The coefficient of x^{m-1} is the same gamma-moment determinant after deleting row zero and column zero, and is positive by (3.1). For $m=1$ the deleted determinant is the empty determinant, equal to one. This proves the theorem. $\square$

Consequence for Q1160. Any negative coefficient in any cycle Hankel minor must occur at a degree strictly between ν and D . The theorem does not establish positivity of those intermediate coefficients.

The highest-coefficient argument also applies to subset rows after replacing the scale $r^{(r-1)}$ in (3.3) by $r^{r/r!}$. The lowest-coefficient argument does not transfer: $[x]s_{\{r,n\}}=1$ has rank one as a Hankel kernel.

An all-parameter subset Hankel countercertificate

Define the related subset minor

$$K_r(x) = \det \begin{pmatrix} s_{r,1}(x) & s_{r,2}(x) & s_{r,3}(x) \\ s_{r,2}(x) & s_{r,3}(x) & s_{r,4}(x) \\ s_{r,3}(x) & s_{r,4}(x) & s_{r,5}(x) \end{pmatrix}. \quad (4.1)$$

This uses Hankel row indices (1,2,3) and column indices (0,1,2).

Theorem 4.1. For every integer $r \geq 3$,

$$K_r(x) = -b^2 \frac{(r-1)(r-2)(r^2 + 11r + 6)}{(r+1)(r+3)(r+2)^2} x^5 + O(x^6), \quad b = \frac{1}{2} \binom{2r}{r}. \quad (4.2)$$

In particular, the coefficientwise Hankel positivity of the subset row polynomials fails for every $r \geq 3$. At each such r , the numerical sequence $(s_{\{r,n\}}(x))$ is also not a Stieltjes moment sequence for all sufficiently small positive x .

Proof. Write $s_{\{r,n\}}(x)/x = 1 + a_n x + O(x^2)$, so $a_n = S(n, 2)$. For $n = 1, \dots, 5$ put

$$a_1 = 0, \quad a_2 = b, \quad a_3 = q, \quad a_4 = e, \quad a_5 = h.$$

Counting the possible two block sizes gives

$$\begin{aligned} b &= \frac{(2r)!}{2(r!)^2}, & q &= \frac{(2r+1)!}{r!(r+1)!}, \\ e &= \frac{(2r+2)!}{r!(r+2)!} + \frac{(2r+2)!}{2((r+1)!)^2}, \\ h &= \frac{(2r+3)!}{r!(r+3)!} + \frac{(2r+3)!}{(r+1)!(r+2)!}. \end{aligned}$$

Factor x from each row of (4.1). Subtract the first row from the second and third rows, then the first column from the second and third columns. The lower-right 2×2 block has first-order part

$$x \begin{pmatrix} q - 2b & e - q - b \\ e - q - b & h - 2q \end{pmatrix}.$$

Every other noncorner entry is $O(x)$, while the corner is 1. Hence coefficients below x^5 in K_r vanish, and

$$[x^5]K_r = (q - 2b)(h - 2q) - (e - q - b)^2. \quad (4.3)$$

The factorial expressions simplify to

$$\frac{q - 2b}{b} = \frac{2r}{r+1}, \quad \frac{h - 2q}{b} = \frac{12(2r+1)}{r+3}, \quad \frac{e - q - b}{b} = \frac{7r+2}{r+2}.$$

Their substitution in (4.3) gives (4.2), because

$$24r(2r+1)(r+2)^2 - (7r+2)^2(r+1)(r+3) = -(r-1)(r-2)(r^2 + 11r + 6).$$

For a fully explicit instance,

and

Appendix C of Deb-Sokal proposes this particular negative-minor mechanism. Formula (4.2) supplies its uniform algebraic proof. This result is a countercertificate for the subset Hankel property, while Q1160 concerns cycle rows.

For an invertible lower triangular array A , its ordinary production matrix is uniquely determined by $AP = \bar{A}$, where $\bar{A}(n, k) = A(n + 1, k)$. The diagonal of each present array is nonzero, so the matrix is well-defined by triangular solution.

For subsets, put $b=S(2,2)$ and $d=S(3,3)$. The corresponding entry is

Thus the triangles themselves cannot be the output matrices of entrywise nonnegative ordinary production matrices. Positive column rescaling only conjugates P by a positive diagonal matrix and cannot remove the negative entry. This does not rule out enlarged-state constructions, different auxiliary arrays, nonstationary networks, or other transformations.

Writing $g_n = c_{\{3,n\}}/x$ or $g_n = s_{\{3,n\}}/x$, the exact quotient recurrences are $g_n = L_n g_{n-1}$, where

24

These follow directly by summing the coefficient recurrences and substituting $f=xg$. They are not generic Hurwitz-stability preservers even on real polynomials with positive coefficients and the required input degree.

For cycles take $n=4$ and $g=50x^2+x+50$. Its two roots have real part $-1/100$. But

$$L_4^C g = 5500x^3 + 522x^2 + 2107x + 250,$$

whose second Hurwitz determinant is

$$522 \cdot 2107 - 5500 \cdot 250 = -275146 < 0.$$

Its Routh first-column signs are $(+,+,-,+)$, certifying two roots in the open right half-plane.

For subsets take $n=6$ and $g=(500x^2+x+500)^2$. Its roots have real part $-1/1000$. The output is

$$L_6^S g = 34000000x^5 + 1355000x^4 + 39004078x^3 + 1555003x^2 + 9002000x + 250000.$$

Here the second Hurwitz determinant is -19576310000 , and the Routh first-column signs are $(+,+,-,+,+,+)$, again certifying two roots in the open right half-plane. These are counterexamples to generic operator preservation, **not** to Q1161: neither input is asserted to be an actual preceding Stirling row.

Exact computations and reusable certificates

Run, with Python 3 and no third-party packages:

```
python3 research/stirling/stirling_research.py
```

The script writes `results.json` and `neville-certificates.json` next to itself.

6.1 Triangle blocks

For each r in $\{2,3,11,12,16,32\}$, both C and S were generated on indices $0,\dots,24$. Every minor of each entire 25×25 block is certified nonnegative by a factorization.

The trivial first row and column split off a 1. On the remaining 24×24 block, bottom-to-top Neville elimination produces

$$A = F_1 F_2 \cdots F_{276} D, \quad F_t = I + m_t E_{i_t, i_t-1}, \quad m_t \geq 0,$$

with D positive diagonal. Every factor is totally nonnegative, and the Cauchy-Binet formula preserves total nonnegativity under multiplication. The exact rational multipliers and diagonal are stored. Reconstructing the block from those factors is separately checked by exact arithmetic during each run. This certifies all finite-block minors without enumerating them.

The $r=2,3$ blocks are controls within the larger source-tested range. The $r=11,12,16,32$ cases add parameter values outside Appendix C's stated range, with a smaller block dimension.

6.2 Third-order roots

Every polynomial $c_{\{3,n\}}(x)/x$ and $s_{\{3,n\}}(x)/x$, $1 \leq n \leq 80$, passed an exact Routh test. The zero at zero is simple because the coefficient of x is positive. The quotient has only roots with strictly negative real parts.

The algorithm forms the Routh rows using integers and removes only positive greatest common divisors. At every stage the entire row differs from the usual rational Routh row by a positive factor, so first-column signs are preserved. All first-column entries were positive. The integer sign sequence, largest pivot bit-length, and a SHA-256 digest of the exact primitive pivots are recorded

for every n ; the generator and deterministic algorithm reproduce every pivot. Singular or zero-pivot cases are explicitly returned as undecided rather than treated as stable. None occurred.

The classical criterion used here is presented with a proof in [Olga Holtz, Hermite-Biehler, Routh-Hurwitz, and total positivity](#). The complete quotients are certified instances; the family for unbounded n is not proved.

6.3 Cycle Hankel blocks

For each r in $\{3,4,11,16,32\}$, all 923 nonempty minors of the 6×6 Hankel block with row and column indices $0, \dots, 5$ were computed in $\mathbb{Z}[x]$ and had nonnegative coefficients. The counts by order are 36,225,400,225,36,1. Thus 4,615 polynomial determinants were checked in total. Bareiss elimination uses exact polynomial division, and the calculated highest and lowest degrees are checked against Theorem 3.1.

The $r=3,4$ calculations are controls. The other three values add parameters outside the source's stated Hankel tests. These are finite checks only.

6.4 Independent controls

The generator reproduces the ordinary Stirling fourth rows $(0,6,11,6,1)$ and $(0,1,7,6,1)$, the exact diagonal factorial formulas, and the third-order third rows. The Routh routine checks a stable cubic, an unstable cubic, and a boundary polynomial with imaginary roots, which must be reported undecided by this strict routine. Polynomial determinants have an independent small symbolic control. The production-matrix formulas and the subset negative coefficient are checked against direct exact calculations over multiple r values.

Next proof targets

The all- r TN2 theorem shifts Q1158 and Q1159 to order-three and higher minors. The direct production-matrix obstruction shows why an auxiliary state space or a different array would be needed for that particular proof method. Positive finite Neville factors suggest seeking closed formulas or inductively positive families of elimination pivots.

For Q1160, the two ends of every determinant polynomial already have strict sign certificates. A remaining negative coefficient, if any, must be internal. A positive proof needs a mechanism controlling cancellations in those coefficients, such as a weighted path model for the Hankel minors. The subset calculation identifies an explicit cancellation mechanism that such a cycle proof must avoid.

For Q1161, exact Routh certificates remove numerical ambiguity on the checked range, but not the induction problem. The ordinary second-order differential recurrence by itself does not automatically preserve Hurwitz stability of arbitrary inputs. The available evidence therefore supports a family-specific invariant, rather than an assumption of generic operator preservation.

No OEIS submission, external publication, durable catalog resolution entry, or edit to the original handoff was made.

Interpolated totally positive sequences and multiplier tests

A compact exact parameter space for the interpolation questions

Write

$$P(z) = \sum_{j=0}^n u_j \binom{z}{j}, \quad L(P)(u) = \sum_{j=0}^n u_j u^j.$$

The source criterion says that P is in $\mathcal{U}_n$ exactly when its leading coefficient is positive and $L(P)$ has every root in $[-1, 0]$. Since positive scaling changes none of the roots under study, normalize $u_n=1$. Then every instance has the form

$$L(P)(u) = \prod_{i=1}^n (u + a_i), \quad 0 \leq a_i \leq 1. \quad (1)$$

This turns the parameter space into a compact cube, modulo permutation. Coefficients vary continuously with the a_i , and the leading coefficient of P stays fixed at $1/n!$. Continuity of polynomial roots as an unordered multiset therefore shows that both r_n and R_n are attained for each fixed degree. Neither supremum needs an escaping sequence of coefficients.

There is also a useful probabilistic/operator representation. Let $\Delta f(z)=f(z+1)-f(z)$, let $(z)_n=z(z-1)\dots(z-n+1)$, and let S be the sum of independent Bernoulli random variables with success probabilities $a_1, \dots, a_n$. The elementary identity $\Delta^k(z)_n = n!/(n-k)! (z)_{n-k}$ gives

$$n!P(z) = \prod_{i=1}^n (1 + a_i \Delta)(z)_n = \mathbb{E}[(z + S)_n]. \quad (2)$$

The factors in (2) are $(1-a_i)+a_i E$, where $E f(z)=f(z+1)$. This is an exact representation of the whole class. An arbitrary distribution on $\{0, \dots, n\}$ would give the larger SNN class; independence imposes the PF numerator condition.

The SNN theorem supplies a strict, though not sharp, imaginary-part bound for the interpolation class. For $n \geq 2$, no member of $\mathcal{U}_n$ can be a positive multiple of M_n : its numerator would be $1+z^n$, which is not a polynomial with all real nonpositive roots. Hence every root of every P in $\mathcal{U}_n$ has $|\operatorname{Im} z| < B_n$. Compactness makes the degree-specific maximum strictly smaller than B_n . This does not imply the desired modulus bound $R_n=n$.

Complete degree-two analysis

By (1), write $L(P)(u)=(u+a)(u+b)$, a, b in $[0, 1]$. Put $s=a+b$, $t=ab$. Then

$$p(z) := 2P(z) = z^2 + (2s-1)z + 2t, \quad (3)$$

where the exact feasible coefficient region is

$$0 \leq s \leq 2, \quad \max(0, s-1) \leq t \leq s^2/4. \quad (4)$$

5.1 Exact nonreal root locus

If $x+iy$, $y \neq 0$, is a root of (3), its conjugate is the other root, and

$$x = \frac{1}{2} - s, \quad y^2 = 2t - x^2.$$

The upper constraint in (4) is precisely

$$\boxed{2(x + 1/2)^2 + 4y^2 \leq 1.} \quad (5)$$

Conversely, for any nonreal point satisfying (5), take $s = 1/2 - x$ and $t = (x^2 + y^2)/2$. The remaining lower constraint in (4) is automatic: it is $t \geq 0$ when $s \leq 1$, and when $s \geq 1$ it reduces to $(x+1)^2 + y^2 \geq 0$. The resulting roots a, b of $v^2 - sv + t$ belong to $[0, 1]$.

Thus (5), away from the real axis, is the exact nonreal root locus. Its maximal imaginary part is $1/2$. The real root locus is $[-2, 1]$. The polynomials with $L(P) = (u+1)(u+a)$ have roots $-1, -2a$, and those with $L(P) = u(u+a)$ have roots $0, 1-2a$; these cover the stated interval. The general real-root bound excludes anything outside it.

5.2 Exact smallest-root extremum

Let

$$r_* = (1 + \sqrt{2})/2, \quad s_* = 1 + 1/\sqrt{2}.$$

For $s \leq s_*$, the smaller root modulus is at most the geometric mean of the two moduli, namely

$$\sqrt{2t} \leq s/\sqrt{2} \leq r_*.$$

For $s \geq s_*$, the discriminant $(2s-1)^2 - 8t$ is nonnegative throughout the feasible region. For fixed s , the smaller modulus increases with t , so it is largest at $t = s^2/4$. There its value is

$$g(s) = s - \frac{1}{2} - \frac{1}{2}\sqrt{2(s-1)^2 - 1}.$$

This function decreases on $[s_*, 2]$: its derivative is $1 - (s-1)/\sqrt{2(s-1)^2 - 1} \leq 0$, with the endpoint interpreted by continuity. Therefore $g(s) \leq g(s_*) = r_*$.

Equality occurs when $a=b=(2+\sqrt{2})/4$, where $p(z)=(z+r_*)^2$. Consequently

$$\boxed{r_2 = (1 + \sqrt{2})/2.}$$

5.3 Largest root and real gaps

For a nonreal root in (5), its modulus is smaller than $r_* < 2$. Real roots lie in $[-2, 1]$, and -2 is attained. Hence $R_2 = 2$.

For real roots the squared gap is $(2s-1)^2 - 8t$. Its maximum at fixed s is attained at the lower endpoint in (4). For $0 \leq s \leq 1$ the bound is $(2s-1)^2 \leq 1$; for $1 \leq s \leq 2$ it is $(2s-3)^2 \leq 1$. Thus the optimal degree-two gap constant is exactly 1.

A proof that $R_3 = 3$

Write $L(P) = \prod (u+a_i)$ for three parameters in $[0, 1]$, and let s_1, s_2, s_3 be their elementary symmetric functions. Then

$$p(z) := 6P(z) = z^3 + (3s_1 - 3)z^2 + (2 - 3s_1 + 6s_2)z + 6s_3. \quad (6)$$

All real roots belong to $[-3, 2]$. It remains to bound a nonreal pair $w, \bar{w}$. Let r be the remaining real root. If $s_3 > 0$, then

$$r|w|^2 = -6s_3, \quad (7)$$

so $r < 0$. Direct substitution in (6) gives

$$p(-s_3) = s_3 F(a_1, a_2, a_3),$$

where

$$F = 4 + 3s_1 - 6s_2 + 3s_3(s_1 - 1) - s_3^2. \quad (8)$$

Here is a finite exact positivity certificate for (8). With $B_i^2(a) = \text{binom}(2, i) a^i (1-a)^{(2-i)}$, the polynomial has the expansion

$$F = \sum_{i,j,k=0}^2 c_{ijk} B_i^2(a_1) B_j^2(a_2) B_k^2(a_3).$$

The coefficients are symmetric under permutation of the three indices; the ten distinct entries are:

Sorted indices	Coefficient
000	4
001	11/2
002	7
011	11/2
012	11/2
022	4
111	29/8
112	5/2
122	1
222	0

Every term is nonnegative on the parameter cube, proving $F \geq 0$. Since

$$p(x) = (x - r)((x - \text{Re } w)^2 + (\text{Im } w)^2),$$

the inequality $p(-s_3) \geq 0$ implies $-s_3 \geq r$. Equations (7)–(8) yield $|w|^2 \leq 6$.

If $s_3 = 0$, take one parameter to be zero and denote the other two by b, c . The polynomial factors as z times a quadratic. If its remaining roots are nonreal, their squared modulus is

$$2 - 3(b + c) + 6bc = 2 - 3b(1 - c) - 3c(1 - b) \leq 2.$$

Thus every nonreal root has modulus at most $\sqrt{6} < 3$. Real roots have modulus at most 3, and $\text{binom}(z+3, 3)$ attains the root -3 . Therefore

$$\boxed{R_3 = 3.}$$

The accompanying code verifies the full Bernstein expansion exactly by its monomial-to-Bernstein coefficient formula. The inequalities in the proof cover all real parameter values, not a parameter grid.

Infinite families giving stronger lower bounds for r_n

For $P(z)=(4z+5)^6$, its sampled-value numerator is

$$N_P(t) = 15625 + 422066t + 1434847t^2 + 963292t^3 + 112567t^4 + 722t^5 + t^6. \quad (9)$$

It has positive coefficients. Its values at

$$-1000, -500, -200, -5, -1, -1/10, 0$$

have signs $+, -, +, -, +, -, +$, respectively. The intermediate value theorem gives six distinct negative roots, exhausting its degree. Thus (9) is an exact all-length PF certificate for $(P(k))$, and every root of P equals $-5/4$.

Exact Sturm certificates also establish $(4z+5)^8$ in U_8 and $(4z+5)^{10}$ in U_{10} . The source's product-closure theorem now gives

$$(4z + 5)^n \in U_n \text{ for every even } n \geq 6.$$

Indeed every integer at least 3 is a sum of 3, 4 and 5. This proves $r_{n \geq 5} \geq 5/4$ for every even $n \geq 6$.

A stronger rational value is obtained from eight exact bases:

$$(40z + 51)^n \in U_n, \quad n = 16, 18, 20, 22, 24, 26, 28, 30.$$

Their complete numerator coefficients and exact root-count certificates are in `shifted-power-family-certificates.json`. Every even integer $n \geq 16$ is a sum of these eight even integers, so product closure proves

$$r_n \geq 51/40 = 1.275 \text{ for every even } n \geq 16.$$

These finite bases have become infinite-family certificates through a structural theorem. They do not establish the requested uniform upper bound.

There is also a useful limitation on this particular construction. If $1 < t < 2$ and $(z+t)^{(2m)}$ belongs to $U_{(2m)}$, the reflected sampled sequence $|k+1-t|^{(2m)}$ is log-concave. Its window $k=0, 1, 2$ gives

$$(2-t)^2 \geq (t-1)(3-t),$$

and therefore $t \leq 2 - 1/\sqrt{2}$, approximately 1.29289. This is an upper bound for these shifted powers, not for general members of U_n .

Exact real-root gap witnesses

Consider the symmetric family

$$P_c(z) = (z^2 + z - c)^2, \quad c \geq 0.$$

Its two distinct real roots have gap $\sqrt{1+4c}$, each with multiplicity two. Direct numerator calculation gives the reciprocal polynomial

$$N_c(t) = c^2 + (4 - 4c - 4c^2)t + (16 + 8c + 6c^2)t^2 + (4 - 4c - 4c^2)t^3 + c^2t^4.$$

For $c > 0$, divide by t^2 and set $v = t + t^{-1}$. The resulting quadratic is

$$c^2v^2 + 4(1 - c - c^2)v + (16 + 8c + 4c^2).$$

Its discriminant is $16(1-2c-5c^2)$. It has both roots in $(-\infty, -2]$ exactly for

$$0 < c \leq c_* = (\sqrt{6} - 1)/5.$$

For this range its vertex lies to the left of -2 , while its value at -2 is $8+16c+16c^2>0$, so the discriminant condition indeed puts both roots there. Each such v yields two negative real values of t . Thus P_c in U_4 throughout this parameter range. The endpoint yields a necessary lower bound

$$D \geq \sqrt{(1 + 4\sqrt{6})/5}.$$

A short stronger certificate is available in degree eight:

$$P(z) = (3z^2 + 3z - 1)^4,$$

whose numerator coefficients, in ascending order, are

$$(1, 616, 77932, 771352, 1566118, 771352, 77932, 616, 1).$$

After dividing by t^4 and substituting $v=t+t^{-1}$, the numerator becomes

$$g(v) = v^4 + 616v^3 + 77928v^2 + 769504v + 1410256.$$

The exact values at $v=-500, -200, -10, -5, -2$ are

$$4598658256, -363370544, 902016, -565439, 178048.$$

Their alternating signs prove four distinct roots strictly below -2 , hence eight distinct negative roots of the numerator. The original polynomial's gap is $\sqrt{7/3}$.

The strongest bounded witness retained here is

$$P(z) = (5z^2 + 5z - 2)^{12} \in U_{24}.$$

Its numerator has positive integer coefficients and 24 distinct real roots, certified by exact Sturm arithmetic. Its two real roots are

$$\frac{-1 - \sqrt{13/5}}{2}, \quad \frac{-1 + \sqrt{13/5}}{2},$$

both of multiplicity 12. Therefore any universal gap constant in Q1443 must satisfy

$$D \geq \sqrt{13/5} \approx 1.61245155.$$

All coefficients and the exact root-count certificate are in `verification-results.json`. This does not contradict the proposed $D < 2$.

The coordinating agent identified a useful ceiling for this family. For even m , if $(z^2+z-c)^m$ in $U_{(2m)}$ and $0 < c < 2$, sampled log-concavity at indices 0, 1 and 2 gives $(2-c)^2 \geq c(6-c)$. Therefore

$$c \leq (5 - \sqrt{17})/2, \quad \text{gap} \leq \sqrt{11 - 2\sqrt{17}} \approx 1.65945.$$

This family cannot approach a gap of 2.

Q1249: a rigorous obstruction to a tempting weakening

The all-remainder condition does not ensure preservation of arbitrary log-concave sequences. A small exact example is

$$A(z) = 2 + 27z + 108z^2 + 108z^3 = 108(z + 1/6)^2(z + 2/3).$$

Every remainder belongs to LP-I: after removing its power of z , the nontrivial tails are $27(1+2z)^2$, $108(1+z)$ and 108 . Let

$$b = (1, 1, 1, 1/2, 1/4, 1/8, \dots).$$

Its successive ratios are $1, 1, 1/2, 1/2, \dots$, so it is log-concave. But the termwise product has generating polynomial

$$2 + 27z + 108z^2 + 54z^3,$$

whose discriminant is -472392 ; it has a nonreal pair. This demonstrates that ordinary log-concavity is too weak to replace the input PF condition in a proof of Q1249.

It is **not a counterexample to Q1249**. The input itself fails PF: its Toeplitz minor with rows $\{0, 1, 2\}$ and columns $\{1, 2, 3\}$ is

$$\det \begin{pmatrix} 1 & 1 & 1/2 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} = -1/2.$$

A separate bounded experiment constructed one exact rational degree-seven A whose eight tails are all real-rooted with nonnegative coefficients. It tested 300 PF inputs with generating functions

$$\frac{\prod_i (1 + \alpha_i z)}{\prod_j (1 - \beta_j z)}, \quad \alpha_i, \beta_j > 0,$$

using deterministic seed 1249, at most three numerator factors and one to four denominator factors. The 300 trials contain 293 distinct input generating functions. Every output passed the exact finite-polynomial PF criterion. Full parameters are recorded. This remains a bounded screen of one candidate, without a sufficiency proof.

Reproduction and limitations

Run from this directory or pass the full paths:

```
python verify_interpolation.py
python additional_certificates.py
```

The scripts use the standard library only. Polynomial coefficients are exact rational numbers or integers. The Sturm procedure removes repeated factors before comparing the distinct real-root count with the squarefree degree. Positive numerator coefficients then exclude positive roots. This decides complete polynomial instances and, through the generating-function criterion, their entire sampled sequences.

The principal generated records are:

- `verification-results.json`: the SNN phase controls, 336 explicit three-term witness controls, and 42 central-gap-family checks.
- `additional-certificate-results.json`: the exact cubic Bernstein certificate, short sign proofs, and the log-concavity proxy obstruction.
- `shifted-power-family-certificates.json`: 11 complete numerator certificates underlying the two infinite shifted-power families.

- `Q1249-bounded-search.json`: the exact degree-seven multiplier candidate and every one of the 300 PF test inputs.
- `shifted-power-results.json`: an earlier bounded reconnaissance over ten even degrees through 64 and five rational shifts; it is not used for the main claims.

No unresolved question is marked proved merely because the listed computations passed. No public submission or durable external catalog update has been made.

Q167: Complete classification in degrees at most four

Date: 2026-10-06. Status: exact structural classification for $n=1,2,3,4$; the general classification remains unresolved. All ties are included, as required by Zhang's association on printed pp. 91–92.

Let $p(x) = \prod_{i=1}^n (1 + r_i x) = \sum_{k=0}^n c_k x^k$, with positive integers r_i . Write an ordering as a string of indices from smallest coefficient to largest. For example, 04123 means $c_0 \leq c_4 \leq c_1 \leq c_2 \leq c_3$.

Classification and explicit witnesses

Degree n	Ordering	One tuple of factors r	Coefficients $(c_0, \dots, c_n)$
1	01	(2)	(1,2)
1	10	(1)	(1,1)
2	012	(3,3)	(1,6,9)
2	021	(1,2)	(1,3,2)
2	201	(1,1)	(1,2,1)
3	0312	(1,1,2)	(1,4,5,2)
3	0321	(1,1,1)	(1,3,3,1)
3	3012	(1,1,1)	(1,3,3,1)
3	3021	(1,1,1)	(1,3,3,1)
3	0132	(1,3,3)	(1,7,15,9)
3	0123	(4,4,4)	(1,12,48,64)
4	04132	(1,1,1,2)	(1,5,9,7,2)
4	04312	(1,1,1,1)	(1,4,6,4,1)
4	40132	(1,1,1,1)	(1,4,6,4,1)
4	40312	(1,1,1,1)	(1,4,6,4,1)
4	04123	(1,1,2,3)	(1,7,17,17,6)
4	01423	(1,2,2,2)	(1,7,18,20,8)
4	01243	(3,3,3,3)	(1,12,54,108,81)
4	01234	(5,5,5,5)	(1,20,150,500,625)

Thus the exact counts are 2,3,6,8. Degree zero, if included, has the unique order 0.

Preliminary inequalities

All c_k are positive. Newton's inequalities for a product of linear factors with positive parameters imply strict ordinary log-concavity at internal indices:

$$c_k^2 \geq \frac{(k+1)(n-k+1)}{(k(n-k))} c_{k-1} c_{k+1} > c_{k-1} c_{k+1}.$$

Consequently the coefficients strictly increase up to their mode and strictly decrease afterwards, allowing only a pair of equal adjacent modes. Sorting their values therefore removes the left or right endpoint of the remaining interval at each step. Nonadjacent equal coefficients can be sorted either way, but they must remain consistent with this endpoint-removal description.

For $r_i \geq 1$ we have $c_0 = 1$, $c_n = \prod r_i \geq 1$, and every internal $c_k \geq \binom{n}{k} > 1$ for $n \geq 2$. Thus c_0 is uniquely minimal unless every $r_i = 1$; in that exceptional case $c_0 = c_n = 1$.

For $n=3$, $c_2 \geq c_1$, with equality only when every $r_i = 1$. Indeed $c_2 - c_1$ is zero at (1,1,1), and its partial derivative with respect to r_i is the sum of the other two factors minus 1, which is positive for $r_i \geq 1$.

For $n=4$, $c_2 > c_1$ always: double-counting gives $2c_2 = \sum_i r_i \sum_{j \neq i} r_j \geq 3c_1$. Also $c_3 \geq c_1$, with equality only at (1,1,1,1): $c_3 - c_1$ vanishes there, while its derivative with respect to any r_i is the sum of the three pair products among the other variables minus 1, at least 2.

Exhaustiveness for $n=1,2,3$

Degree one has two possible orders, both exhibited.

For degree two, $c_1 > c_0$; the only additional endpoint tie $c_0 = c_2$ occurs at $(1,1)$. Otherwise c_2 can lie on either side of c_1 . This gives exactly the three listed orders.

For degree three with some $r_i > 1$, c_0 is uniquely minimal and $c_1 < c_2$. The last coefficient c_3 can lie below c_1 , between c_1 and c_2 , or above c_2 , giving 0312, 0132, 0123. At $(1,1,1)$, both endpoint and middle ties occur, giving 0312, 0321, 3012, 3021. Their union is the six orders listed.

The degree-four obstruction

Suppose $n=4$, some factor exceeds 1, and $c_2 \geq c_3$. Then

$$c_4 < c_1. \quad (*)$$

This uses integrality. Let s be the number of factors that are at least 2.

- If $s=1$, the tuple is $(1,1,1,a)$ up to permutation and $c_4 = a < c_1 = a+3$.
- If $s=2$, the tuple is $(1,1,a,b)$, where $a,b \geq 2$. Then $c_2 - c_3 = 1 + a + b - ab$. The hypothesis implies $ab \leq a + b + 1 < a + b + 2 = c_1$, proving (*).
- If $s=3$, the tuple is $(1,a,b,c)$, where $a,b,c \geq 2$. Then $c_3 - c_2 = abc - a - b - c$. This is at least $8 - 2 - 2 - 2 = 2$, because it is increasing in each variable on $[2, \infty)^3$. Hence $c_2 \geq c_3$ is impossible.
- If $s=4$, all $r_i \geq 2$. Double-counting extensions of pairs gives $3c_3 = \sum_{|S|=2} \prod_{i \in S} r_i \sum_{j \notin S} r_j \geq 4c_2$, so again $c_3 > c_2$.

That proves (*).

For a tuple other than $(1,1,1,1)$, the inequalities $c_0 < c_1 < c_2$ and $c_1 < c_3$, together with strict log-concavity, leave only six endpoint-removal orders:

04132, 01432, 04123, 01423, 01243, 01234.

The order 01432 would require $c_4 \geq c_1$ and $c_2 \geq c_3$, contradicting (*). The remaining five orders all have explicit witnesses above. The exceptional all-ones tuple contributes 04312, 40132, 40312 in addition to 04132. Thus there are exactly eight orders.

A finite search over bounded factors is unnecessary for this exhaustiveness claim: the case analysis covers every positive integer tuple, regardless of factor size. The source only reports a general empirical growth guess; no novelty claim is made for this small-degree classification.

General obstruction available in every degree

For positive integers r_i , let $s = \#\{i: r_i \geq 2\}$. Then

$$c_{k+1}/c_k \geq 2(s-k)/(k+1) \quad \text{whenever } 0 \leq k < s.$$

Indeed, for each k -set S at least $s-k$ factors outside S are at least 2, so

$$(k+1)c_{k+1} = \sum_{|S|=k} r_S \sum_{j \notin S} r_j \geq 2(s-k)c_k.$$

Including the $n-s$ factors merely equal to at least 1 gives the stronger universal form

$$(k+1)c_{k+1} \geq (n+s-2k)c_k \quad \text{for } k \leq s,$$

because among the $n-k$ outside factors at least $s-k$ contribute an additional unit. These inequalities supply immediate forbidden coefficient orders based on the number of nonunit factors. They do not by themselves classify all orders.

Sources

- Yufei Zhang, Some Combinatorial Problems Involving Total Non-Negativity and Unimodality, Notre Dame, 2025, pp. 91–98, especially Question 5.2.6 and its sorting/tie convention. <https://ndownloader.figshare.com/files/54098216>. The classification here is an independent deduction from that setup.

- Newton's inequalities are classical; the only use here is the displayed strict log-concavity consequence for positive elementary symmetric functions. The proofs of the new small-degree obstructions are given in full above.

`verify_q167.py` independently checks every listed witness and enumerates every permutation compatible with its exact coefficients; its finite controls are in `q167-controls.json`.

Eulerian recognition, atoms, pattern intervals, and random graphs

Research results for B3, 6 October 2026. This report covers Q1262–Q1263, Q1366, Q2212 and Q2400, and supplies controls for Q183 and Q1454. It accompanies the other mathematical reports and the exact positivity toolkit.

Q1262: a finite recognition theorem for every possible length

For a positive integer vector $s = (s_1, \dots, s_L)$, use the inversion-sequence definition

$$E_s(x) = \sum_{0 \leq e_i < s_i} x^{\text{asc}(e)}, \quad \text{asc}(e) = \#\{0 \leq i < L : e_i/s_i < e_{i+1}/s_{i+1}\},$$

where $e_0 = 0, s_0 = 1$. The equivalence of this definition with the lecture-hall Ehrhart definition is established in the literature cited by Deligeorgaki-Han-Solus. Their Proposition 25 already uses factorization of $E_s(1)$ and removal of repeated entries equal to one to give an exact exclusion. The recognition theorem below makes that method general and explicit; it is not a claim to have discovered the unit-removal argument.

Theorem. Let $p(x)$ have nonnegative integer coefficients and constant term one, and put $N = p(1)$. If $N > 1$, then $p = E_s$ for some positive integer vector of arbitrary length if and only if it equals one of the finitely many E_s obtained as follows:

1. Choose any ordered factorization $N = t_1 \cdots t_q$ with every $t_i \geq 2$.
2. Insert either no entry or one entry equal to one in each of the $q - 1$ internal gaps.

Here $q \leq \Omega(N)$, the number of prime factors counted with multiplicity, and the resulting length is at most $2\Omega(N) - 1$. If $N = 1$, the only possible polynomial is one, represented by any all-ones vector.

Proof. The number of inversion sequences is $E_s(1) = \prod_i s_i$. All nonunit entries therefore form an ordered factorization of N . A coordinate with $s_i = 1$ forces $e_i = 0$. Leading or trailing entries equal to one add no ascents and can be removed. A consecutive block of ones can be reduced to a single one: every removed coordinate and its neighboring unit coordinate both have ratio zero, so the ascent statistic is unchanged. These deletions give an ascent-preserving bijection, leaving only the stated choices. Conversely every stated choice is a valid positive vector. The length and factor-count bounds follow directly. $\square$

Thus a negative output of `eulerian_recognition.py` is a complete exclusion for that polynomial, covering all lengths. It is not a fixed-dimension nonmatch. This is an effective decision procedure; a structural characterization of all (m, r) is still missing.

A small complete exclusion

For the uncolored multiplicities $(2, 3)$,

$$A_{(2,3)}^{(1,1)}(x) = 1 + 6x + 3x^2, \quad A_{(2,3)}^{(1,1)}(1) = 10.$$

The canonical candidates are $(10), (2, 5), (5, 2), (2, 1, 5), (5, 1, 2)$. Their distinct polynomials are

$$1 + 9x, \quad 1 + 7x + 2x^2, \quad 1 + 5x + 4x^2.$$

None is the required polynomial. This proves an exclusion for every positive integer vector s , independently of a length bound chosen by the computation. The source's $(3, 3)$ exclusion is also reproduced.

An infinite family from a Pell equation

A unit separator gives a product identity:

$$E_{(a,1,b)}(x) = E_{(a)}(x)E_{(b)}(x) = (1 + (a-1)x)(1 + (b-1)x). \quad (1.1)$$

Indeed, the middle inversion coordinate is zero. Its incoming transition is never an ascent, and its outgoing transition is exactly the initial ascent for the second block. More generally $E_{(s,1,t)} = E_s E_t$ for arbitrary positive vectors s, t .

For two uncolored symbols with multiplicities $(2, b)$,

$$A_{(2,b)}^{(1,1)}(x) = 1 + 2bx + \binom{b}{2}x^2.$$

Whenever $b(b+1)/2 = t^2$,

$$1 + 2bx + \binom{b}{2}x^2 = (1 + (b-t)x)(1 + (b+t)x),$$

and consequently

$$\boxed{A_{(2,b)}^{(1,1)} = E_{(b-t+1, 1, b+t+1)}} \quad (1.2)$$

Both entries are positive because $t \leq b$ for $b \geq 1$. The positive solutions form an infinite family: starting from $(b, t) = (1, 1)$, the recurrence

$$(b, t) \mapsto (3b + 4t + 1, 2b + 3t + 1)$$

preserves $(2b+1)^2 - 8t^2 = 1$ and strictly increases b . Thus the first nontrivial examples are

b	t	A representing vector s
8	6	(3, 1, 15)
49	35	(15, 1, 85)
288	204	(85, 1, 493)
1681	1189	(493, 1, 2871)

No general exclusion is inferred for values outside this family. For example $(2, 2)$ is represented by $(2, 3)$ through a different mechanism.

Exact finite recognition results

The script generates colored coefficients from the supplied Ehrhart identity and compares complete polynomials. The refined inversion-sequence dynamic program uses integer cross multiplication to compare ratios and was independently checked against literal enumeration for all 120 vectors of length at most four with entries in $\{1, 2, 3\}$.

For $1 \leq a \leq 6$, $a \leq b \leq 8$, retaining exactly the cases with $\binom{a+b}{a} \leq 500$, all 28 uncolored two-symbol polynomials were completely decided. Ten are s-Eulerian: the eight cases $a = 1$, plus $(2, 2)$ and $(2, 8)$. The other eighteen are excluded for every possible length. The complete coefficients, witness vectors, and candidate counts are in `eulerian_results.json`.

Q1263: exact symmetric decompositions

For a polynomial p of its actual degree d , the decomposition $p = a + xb$ is obtained exactly from

$$b(x) = \frac{x^d p(1/x) - p(x)}{1 - x}, \quad a(x) = p(x) - xb(x).$$

Successively removing the lowest-degree term in the symmetric basis $x^i(1+x)^{d-2i}$ then gives the two gamma-vectors without numerical fitting. This is implemented in both the recognition script and the general toolkit.

As an elementary closed formula, take two distinct symbols with color counts $u, v \geq 2$. Then

$$A_{(1,1)}^{(u,v)}(x) = 1 + (uv + u + v - 2)x + (u - 1)(v - 1)x^2.$$

Its actual degree is two, and its two gamma-vectors are

$$\gamma(a) = (1, 2u + 2v - 4), \quad \gamma(b) = ((u - 1)(v - 1) - 1).$$

These are nonnegative on the stated parameter range. They give complete degree-two formulas, but no general orbit action or interpretation of the two vectors for arbitrary multiplicities is supplied. Q1263 remains unresolved in its requested general form.

Q2212: exhaustive finite coverage of every numerical monoid

Two elementary reductions make an exact finite screen exhaustive over monoids, rather than an arbitrary list of generators.

Atom inheritance. If $A \subset H$ contains zero and $A = B + C$ with both factors containing zero, then $B, C \subset A$. Therefore A is an atom in the reduced power monoid of H if and only if it is an atom in that of $\mathbb{N}_0$. Its atomicity is independent of the surrounding numerical monoid.

Truncation characterization. A set $T \subseteq \{0, \dots, \ell\}$ containing zero occurs as $H \cap [0, \ell]$ for a numerical monoid H if and only if

$$a, b \in T, \quad a + b \leq \ell \quad \implies \quad a + b \in T.$$

Necessity is immediate. For sufficiency, take $H = T \cup \{\ell + 1, \ell + 2, \dots\}$. Sums at most ℓ are covered by the condition, and all larger sums lie in the cofinite tail. This is a numerical monoid with the required truncation.

The C++ screen enumerates all such T . Independently, it enumerates every pair of nonidentity finite factors B, C containing zero with $\max B + \max C \leq \ell$, and marks their sumsets as decomposable. Restricting the remaining subsets to each T gives the required atom counts. The output retains exactly cardinalities $0, \dots, \ell$, as required by the source, and maximum **at most** ℓ .

Result. Every numerical monoid satisfies Q2212 for every cutoff $0 \leq \ell \leq 22$. In fact every counted row in this finite range is log-concave.

Cutoff ℓ	Distinct numerical-monoid truncations	Nonempty subsets evaluated across those truncations
12	171	16,758
16	785	306,484
18	1,655	1,282,272
20	3,516	5,315,701
22	7,257	21,814,098

At cutoff 22, the sunset stage checked 20,972,544 unordered factor pairs. Arithmetic is exact; the chosen cutoff keeps every counter and its squared inequality within unsigned 64-bit range. An independent Python implementation enumerated factors as literal sets for all 502 nonempty candidate subsets through cutoff eight, and matched all 110 monoid rows in those cutoffs. It shares neither the bitwise sunset operation nor the factor-pair enumeration scheme with the C++ implementation.

These are complete finite results, with every row retained in the accompanying JSONL files. They do not establish the conjecture for $\ell \geq 23$.

A source-status correction

The handoff follows Tringali’s survey in describing the cited special-case asymptotic result as unimodality for almost all cutoffs. The cited Aggarwal-Gotti-Lu preprint’s Corollary 5.3 actually states that asymptotically almost all **positions within a row** satisfy the appropriate direction of inequality. That is a different quantifier. The retrieved preprint has only its December 2024 version; its abstract also describes each row as “almost unimodal.” The stronger almost-all-cutoffs interpretation should not be treated as established by that cited corollary. This report does not attempt to re-prove the asymptotic result.

Q1366: exact classical-pattern interval controls

The screen constructs every principal downset of every permutation of length at most seven by recursive deletion and standardization. A set records distinct standardized permutations, so repeated occurrences do not inflate any rank. Literal position-subset enumeration independently verifies every downset through top length five.

For each top permutation τ , every lower pattern $\sigma \leq \tau$ is used as an interval endpoint. Counting each intermediate distinct permutation by its length gives the complete interval rank vector.

All 200,523 intervals with nonempty lower endpoint and top length at most seven were unimodal. There were 206,436 intervals when the empty lower permutation was also included. These totals include all smaller top lengths; the detailed per-length counts are in `pattern_interval_results.json`.

For calibration, the interval $[1, 13524]$ has rank vector

$$(1, 2, 5, 5, 1).$$

It is unimodal and fails log-concavity because $2^2 < 1 \cdot 5$. This is an exact witness against an unrequested stronger property, and provides no counterexample to Q1366. The full all-interval conjecture remains unresolved.

Q2400: a rigorous growing prefix with high probability

The requested assertion concerns every coefficient of a sampled graph simultaneously, in a limit with fixed d . The following theorem establishes a growing initial part of that assertion. It leaves the remainder, including the region near the independence number, unresolved.

Deterministic lemma. Let G have $n \geq 2$ vertices and m edges. For $1 \leq k \leq n - 1$, define

$$\epsilon_k = \frac{mk(k-1)}{n(n-1)}, \quad T_k = \frac{n+1}{(k+1)(n-k+1)}.$$

If $2\epsilon_k \leq T_k$, then

$$x_k(G)^2 \geq x_{k-1}(G)x_{k+1}(G). \tag{5.1}$$

Proof. Every non-independent k -set contains an edge. A union bound, expressed as an exact counting inequality, gives

$$x_k \geq \binom{n}{k} - m \binom{n-2}{k-2} = \binom{n}{k} (1 - \epsilon_k).$$

Also $x_{k-1} \leq \binom{n}{k-1}$ and $x_{k+1} \leq \binom{n}{k+1}$. The hypothesis implies $0 \leq \epsilon_k \leq 1/2$, and

$$(1 - \epsilon_k)^2 \geq 1 - 2\epsilon_k \geq 1 - T_k = \frac{\binom{n}{k-1} \binom{n}{k+1}}{\binom{n}{k}^2}.$$

Squaring the nonnegative lower bound proves (5.1). $\square$

Corollary. Fix $d > 0$ and take $n \geq \max(2, d)$ with $G \sim G(n, d/n)$. Put

$$K = \min \left(n-1, \left\lfloor \left(\frac{n-1}{2d} \right)^{1/3} \right\rfloor \right).$$

With probability at least $1 - \exp(-d(n-1)/6)$, (5.1) holds simultaneously for every $1 \leq k \leq K$.

Proof. The edge count has mean $\mu = d(n-1)/2$. The standard binomial Chernoff bound gives $\Pr[m > dn] \leq \Pr[m > 2\mu] \leq e^{-\mu/3}$. On $m \leq dn$,

$$2\epsilon_k(k+1) \leq \frac{2d(k^3 - k)}{n-1} \leq 1$$

for every indicated k , while $T_k \geq 1/(k+1)$. The deterministic lemma applies to all those indices at once, without a union bound over k . $\square$

For $d = 0$ the graph is empty and the full coefficient sequence is binomial, hence log-concave. For fixed positive d , the corollary proves a prefix of order $n^{1/3}$ with exponentially high probability. It neither proves nor refutes Q2400's complete-row assertion.

An exact expectation identity

For $q = 1 - d/n$ and arbitrary a, b , the exact mixed moment is

$$\mathbb{E}[x_a x_b] = \sum_j \frac{n! q^{\binom{a}{2} + \binom{b}{2} - \binom{j}{2}}}{j!(a-j)!(b-j)!(n-a-b+j)!},$$

where the sum contains only indices for which all factorials are defined. This counts ordered pairs of independent sets by their intersection size.

Taking $(a, b) = (k, k)$ and $(k-1, k+1)$ shows, term by term, that $\mathbb{E}[x_k^2 - x_{k-1}x_{k+1}] > 0$ for $0 < q \leq 1$ and $1 \leq k \leq n$. For $j \leq k-1$ the second summand divided by the first is $q(k-j)/(k+1-j) < 1$; the $j = k$ term occurs only in the first sum. This concerns the expectation of the defect, and is stronger than comparing only expected coefficients, but still does not imply that the defect is nonnegative with high probability. The growing-prefix corollary above is the part that makes an actual probability statement about sampled graphs.

Q183 and Q1454: structural calibrations and remaining gaps

For Q183, every board with at most two nonempty rows has a non-nesting rook polynomial $1 + (a+b)x + cx^2$, where a, b are the row lengths and $0 \leq c \leq ab$. Its discriminant satisfies

$$(a + b)^2 - 4c \geq (a - b)^2 \geq 0.$$

Thus those instances are real-rooted, and automatically avoid the three-row forbidden board. Transposition gives the corresponding two-column family. This elementary subclass does not handle general forbidden-board avoidance, and no new general resolution of Q183 is asserted.

For Q1454, the Catalan control is an actual infinite certificate:

$$C_n = \frac{1}{2\pi} \int_0^4 x^n \sqrt{\frac{4-x}{x}} dx.$$

The integral is finite even at zero and equals $\frac{1}{n+1} \binom{2n}{n}$ by the beta integral. The shifted atomic sequence from the handoff is also retained with its correct shift:

$$b_j = \frac{1}{8}6^j + \frac{1}{4}4^j + \frac{5}{8}2^j.$$

It is the moment sequence of $\frac{1}{8}\delta_6 + \frac{1}{4}\delta_4 + \frac{5}{8}\delta_2$. Its Hankel rank is exactly three. The toolkit verifies the Catalan blocks and correctly detects the singular atomic blocks. These reproduce known certificates; they do not settle any previously unresolved forbidden pattern. In particular no extrapolated 1324 terms are used.

Sources

- Deligeorgaki-Han-Solus, Colored Multiset Eulerian Polynomials, arXiv:2407.12076, Proposition 25 and Questions 22, 26: <https://arxiv.org/pdf/2407.12076> .
- Tringali, Power monoids and their arithmetic: a survey, arXiv:2602.15754v3, Conjecture 5.4: <https://arxiv.org/html/2602.15754v3> .
- Aggarwal-Gotti-Lu, On primality and atomicity of numerical power monoids, arXiv:2412.05857v1, Corollary 5.3: <https://arxiv.org/html/2412.05857v1> .
- McNamara-Steingrímsson, On the topology of the permutation pattern poset, arXiv:1305.5569: <https://arxiv.org/abs/1305.5569> .
- Vatter, An assortment of problems in permutation patterns, arXiv:2602.16355, Conjecture 2.2 and Question 2.7: <https://arxiv.org/html/2602.16355> .
- Heilman, Independent Sets of Random Trees and of Sparse Random Graphs, arXiv:2006.04756, Question 1.21: <https://arxiv.org/pdf/2006.04756> .
- Jal et al., Rook matroids and log-concavity, arXiv:2410.00127: <https://arxiv.org/abs/2410.00127> .
- Bostan-Elvey Price-Guttmann-Maillard, Stieltjes moment sequences for pattern-avoiding permutations, arXiv:2001.00393: <https://arxiv.org/abs/2001.00393> .

Source questions are attributed above. The elementary deductions and exact calculations are provided with their proofs and scopes, without asserting publication priority.

Exact positivity diagnostic and verification

The delivered `positivity.py` is a standalone implementation of the handoff's proposed diagnostic. It accepts a JSON object or a list of objects and emits structured exact results. It is ready to use independently of the unavailable original `oeis.py` repository.

All scalar input values are integers or rational strings; floating-point values are rejected. Polynomial coefficients are ordered from constant term upward. Metadata is retained, and every bounded-minor result records its row range, column range, maximum order, budget and number of checked minors. A failure includes a concrete exact witness.

Supported inputs

Input kind	Main capabilities	Scope rule
finite-row	Unimodality, log-concavity, log-convexity, internal zeros, ULC, optional Toeplitz minors	Actual complete coefficient row; ULC requires explicit order and indices $0, \dots, N$
prefix	Local inequalities, fully known Hankel PSD blocks, optional Toeplitz window	Unknown entries are never set to zero
matrix	Bounded minors, optional exact PSD	Finite matrix scope is explicit
polynomial-matrix	Exact polynomial minors and negative coefficient witnesses	Values at sampled parameters are not substituted for coefficient signs
polynomial	Exact real-root counting, finite PF criterion, weak half-plane stability, gamma conversion	The entire specified polynomial is certified
interpolated-polynomial	Rational numerator and PF criterion for $(P(k))_{k \geq 0}$	A passing numerator criterion certifies the infinite sampled sequence

Typical use is:

```
python positivity.py research/toolkit/examples.json --out example_results.json
```

The README in the archive specifies every input option and all reproduction commands.

Boundary-safe exact methods

Real roots. A rational Sturm chain counts distinct roots of the squarefree part. Multiplicities and roots at zero are retained separately. For nonnegative polynomial coefficients, real nonpositive roots are exactly the finite PF property.

Weak half-plane stability. For a nonzero real polynomial p with positive leading coefficient, form $p(z+t)$. Every root of p lies in $\operatorname{Re} z \leq 0$ exactly when $p(z+t)$ is strictly Hurwitz stable for every sufficiently small positive t . By the Hurwitz criterion, this is equivalent to every leading Hurwitz determinant, regarded as a polynomial in t , having positive first nonzero coefficient. The diagnostic computes those polynomials exactly. This handles repeated imaginary-axis roots, where a strict test on p alone would fail. The default symbolic degree budget is twelve and is explicit in the output.

Positive semidefiniteness. Exact symmetric elimination handles positive pivots, zero rows, and zero diagonal entries with nonzero off-diagonal entries. A negative result records an actual negative principal minor. It does not infer semidefiniteness from only the leading principal minors.

Moment prefixes. Both $H_0 = (a_{i+j})$ and $H_1 = (a_{i+j+1})$ are tested at the largest sizes entirely known from the prefix. Passing these necessary finite constraints is reported as finite support, without a claim of an infinite representing measure or an extension theorem.

Interpolation. The numerator is computed exactly as

$$[z^j]N_P(z) = \sum_{i=0}^j (-1)^i \binom{n+1}{i} P(j-i), \quad n = \deg P.$$

Nonnegative coefficients and real nonpositive zeros of N_P establish the full sampled-sequence PF property using the cited generating-function criterion.

Independent verification

The toolkit controls include 729 symmetric 3×3 matrices, comparing the PSD procedure against every principal minor; 15 polynomial determinants checked against a separate permutation expansion; 150 constructed real-rooted and non-real-rooted polynomial cases; and 14 half-plane cases including repeated imaginary roots. All passed.

The Catalan prefix gave PSD blocks of sizes eight and seven with full rank. A known three-atom measure gave rank three in both fully known Hankel blocks. Deliberately insufficient prefixes returned insufficient-data rather than a false pass.

The main proofs received separate audits. Q165's rational identities and coefficient formulas were checked by both a symbolic system and an independent implementation using exact fractions; the formal substitution and the analytic implication were reviewed separately. Q166's difference-constraint closure, endpoints and weighted four-functions application were checked directly, with independent composition and cut-set counts as controls. Q325's phase ordering, centered-integral inequality, sharpness and equality cases were checked independently.

The stronger interpolation examples were independently reconstructed and checked with exact root counting, including all eleven bases underlying the two infinite smallest-root families, the degree-24 gap example, and all 27 tensor Bernstein coefficients used for $R_3 = 3$.

Most useful remaining proof targets

The next work should focus on a short list of concrete missing statements.

1. **Q1158 and Q1159:** strengthen total nonnegativity of order two to a uniform factorization, planar network, or explicit higher-minor formula. Exact Neville factors exist for the computed blocks, but a symbolic description valid for arbitrary size is missing. The naive ordinary production matrices have negative entries, so their signs do not directly give the required certificate.
2. **Q1160:** interpret the interior coefficients of the cycle Hankel determinants. The two extremes are already positive by explicit moment arguments; any counterexample would have to occur strictly between the proved degrees. The related subset negative minor identifies a precise feature a cycle-specific proof must use.
3. **Q164:** upgrade the interval-constraint four-functions argument to binomial-normalized ultra-log-concavity, and address circular intervals for cycles. Ordinary log-concavity alone is insufficient. The all-length success for Q165 also suggests investigating whether richer interval recurrences have positive cross-product certificates.
4. **Q1441-Q1443:** use the compact cube parameterization and product closure to sharpen the uniform extremal bounds. Fixed-degree compactness gives existence of extrema, but it supplies no bound independent of degree. The explicit lower examples constrain what a proposed bound can be.
5. **Q1262:** replace finite volume-factorization recognition by structural criteria for colored multiplicities. The unit-separator product identity and Pell family give concrete constructive cases, while exact all-length exclusions can rule out overly broad conjectures.
6. **Q2400:** control adjacent independence-count ratios beyond the first $O(n^{1/3})$ coefficients, especially near the independence number. The positive expectation identity is not a substitute

for concentration of those ratios or defects.

These targets are stronger than extending a passing computational cutoff. The archive preserves the existing exact computations so that subsequent work can start from the proved reductions and retained witnesses.